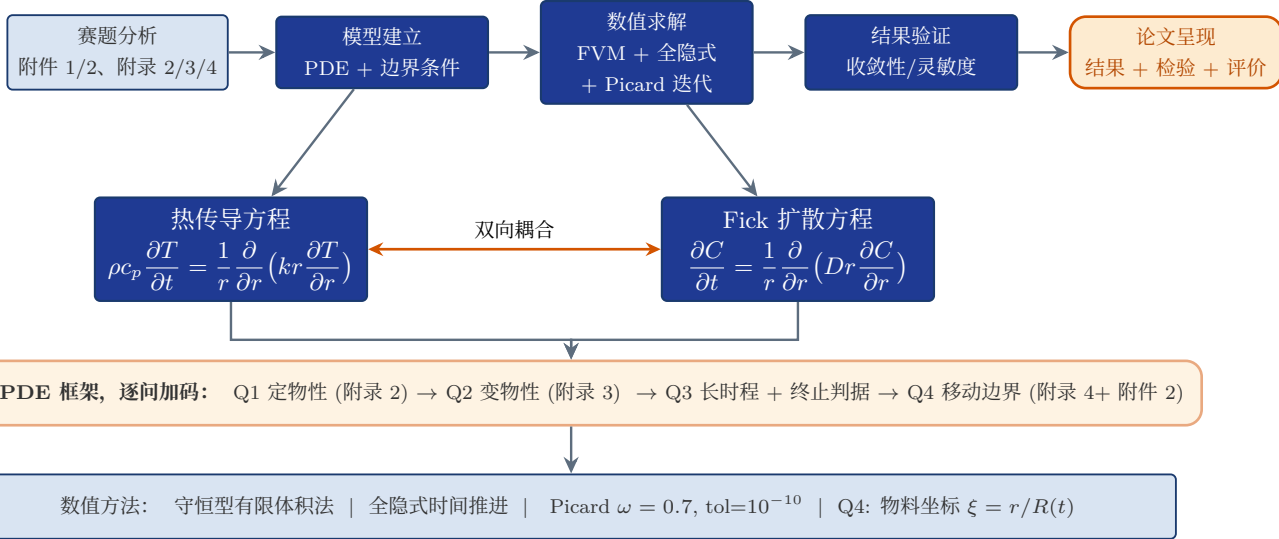
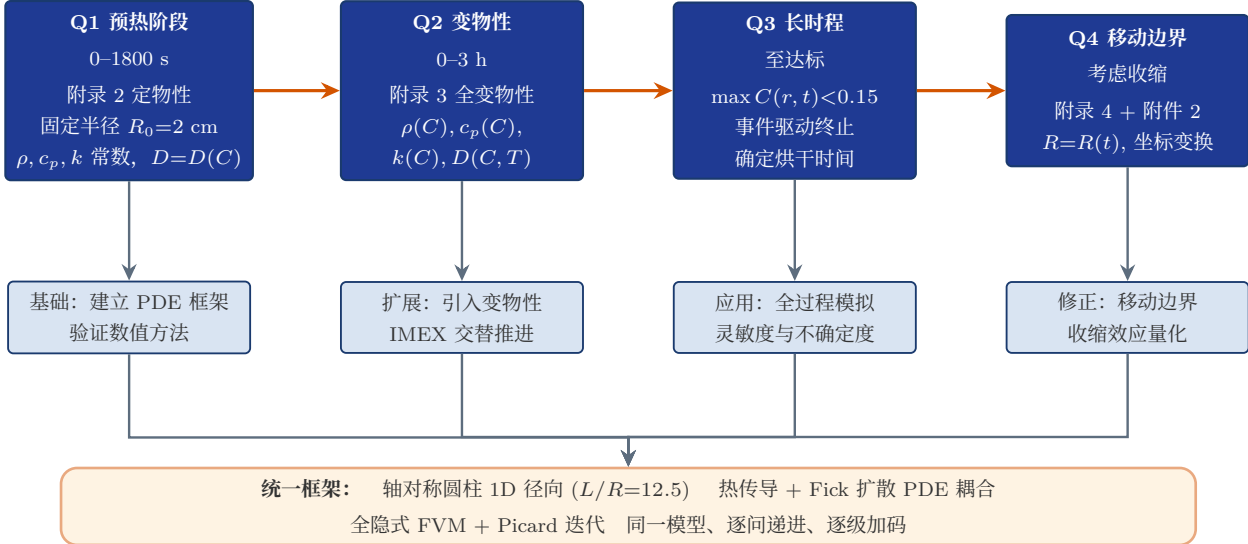
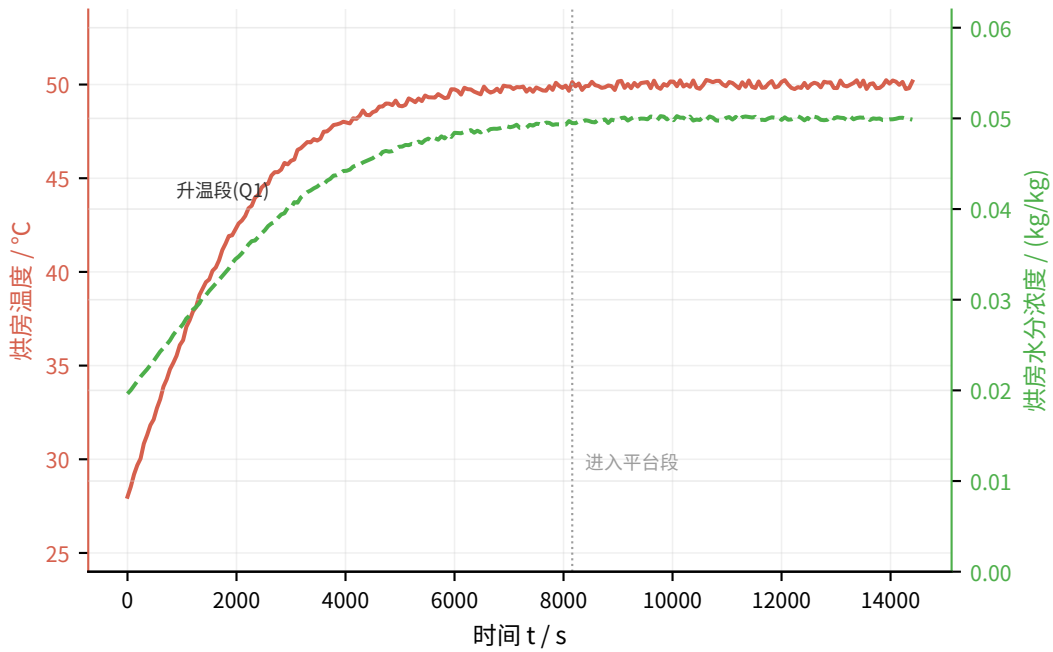


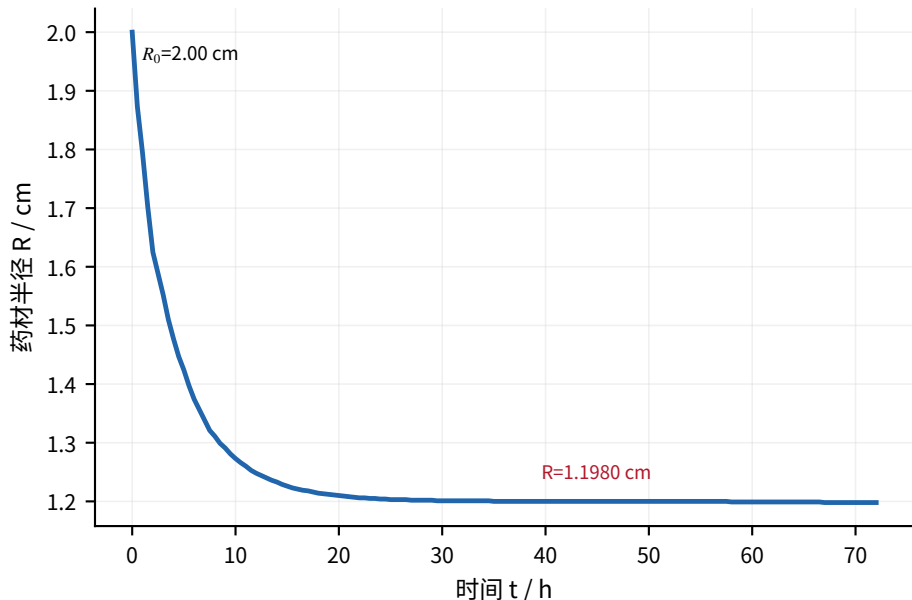




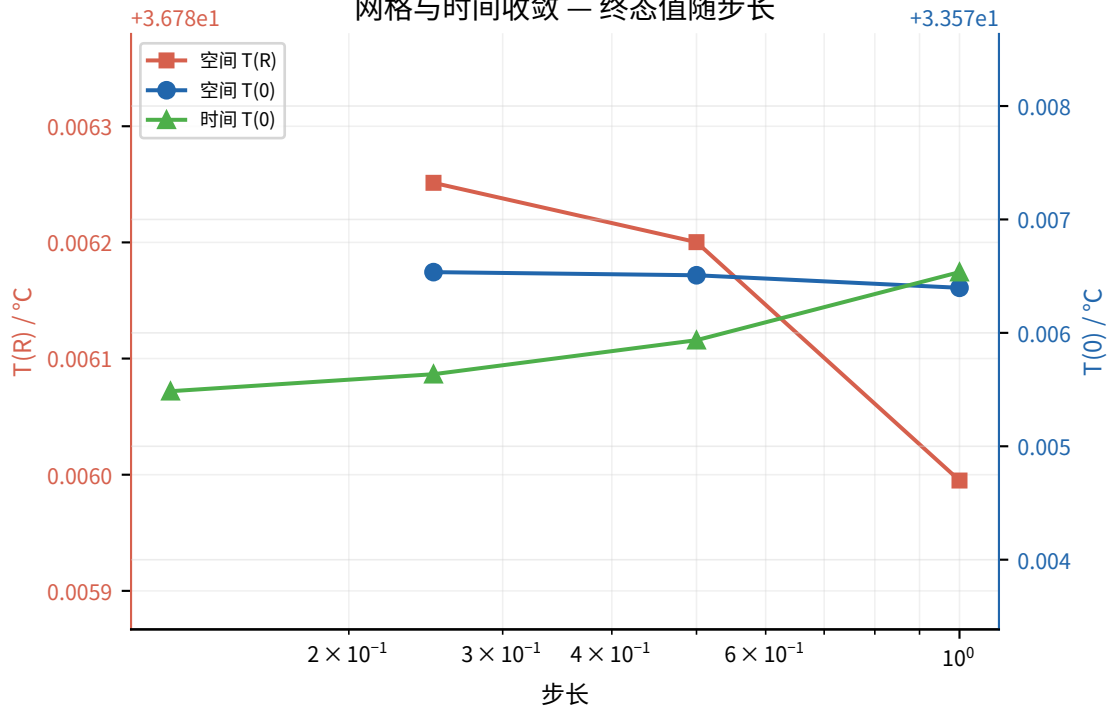
烘房环境条件时序 (附件1)



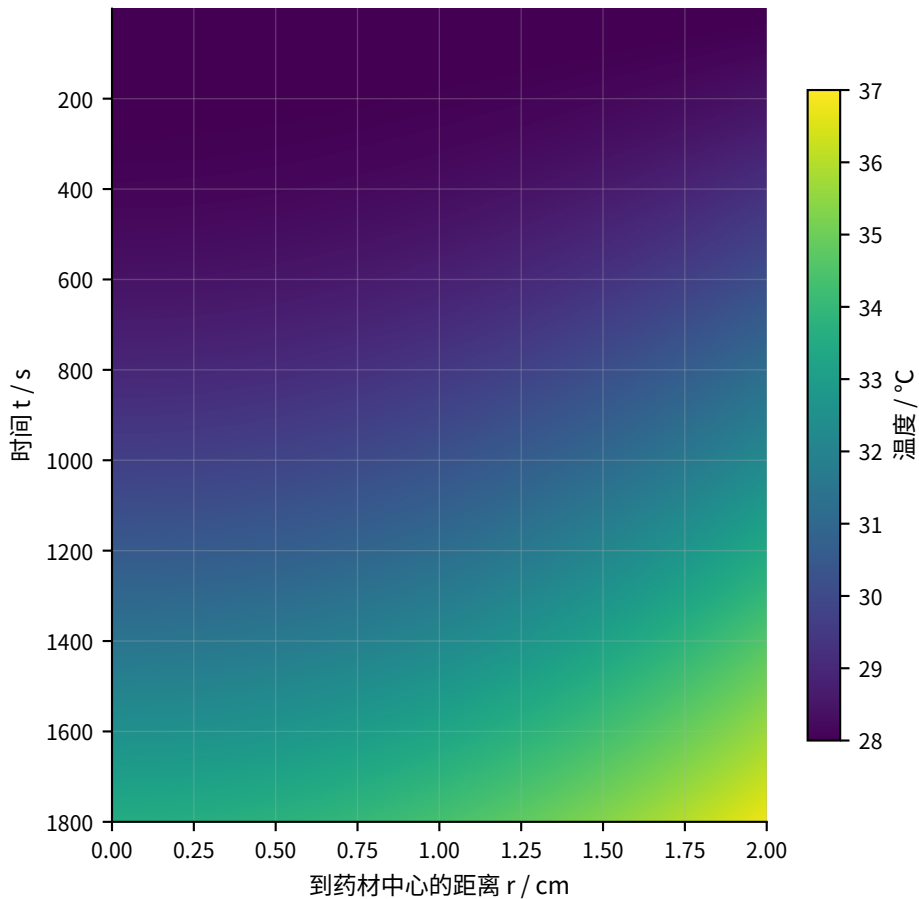
药材半径随干燥时间收缩（附件2）



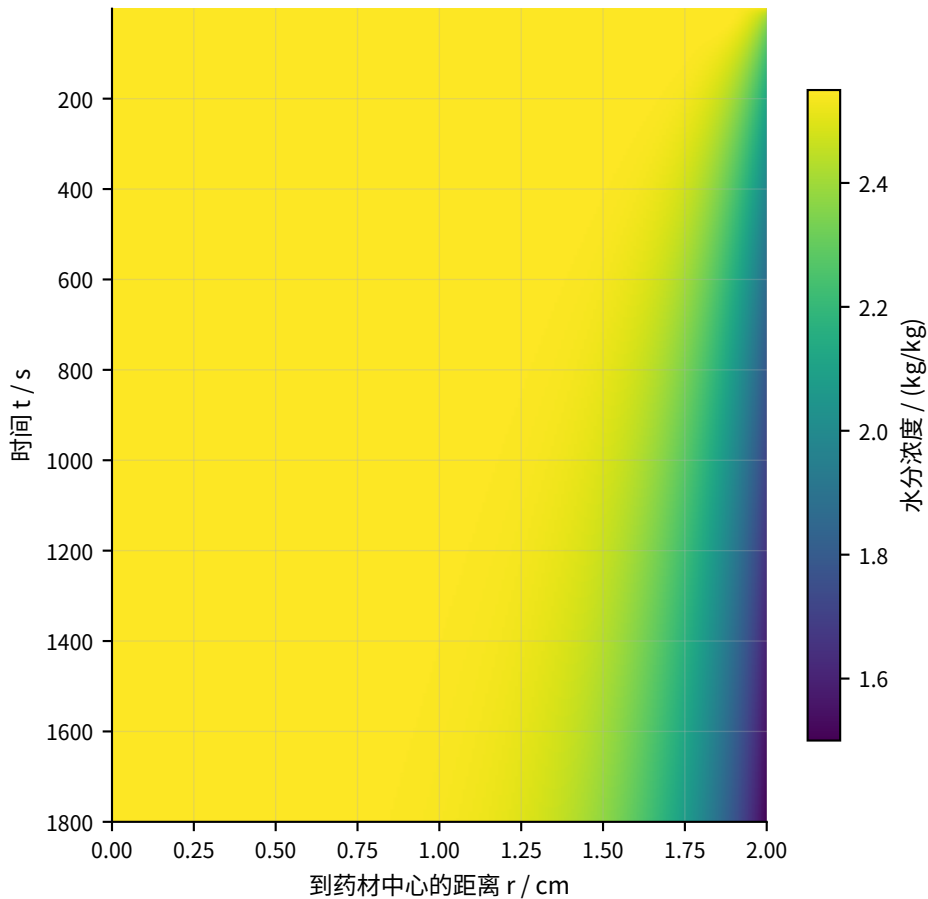
网格与时间收敛 — 终态值随步长



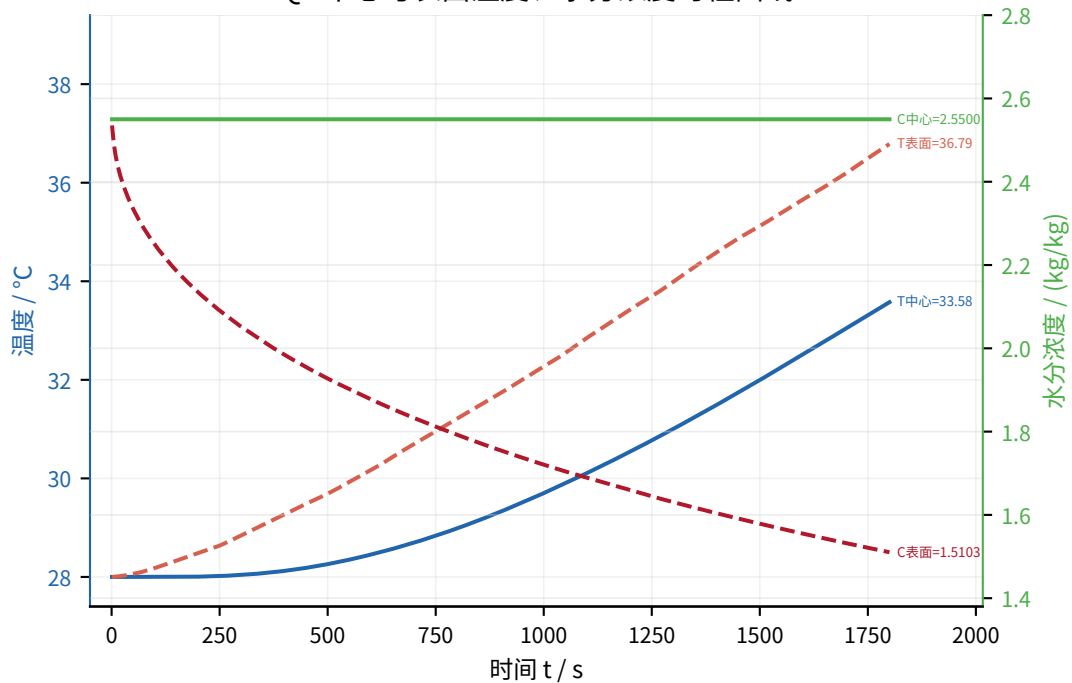
Q1 温度场 $T(r,t)$ 时空分布



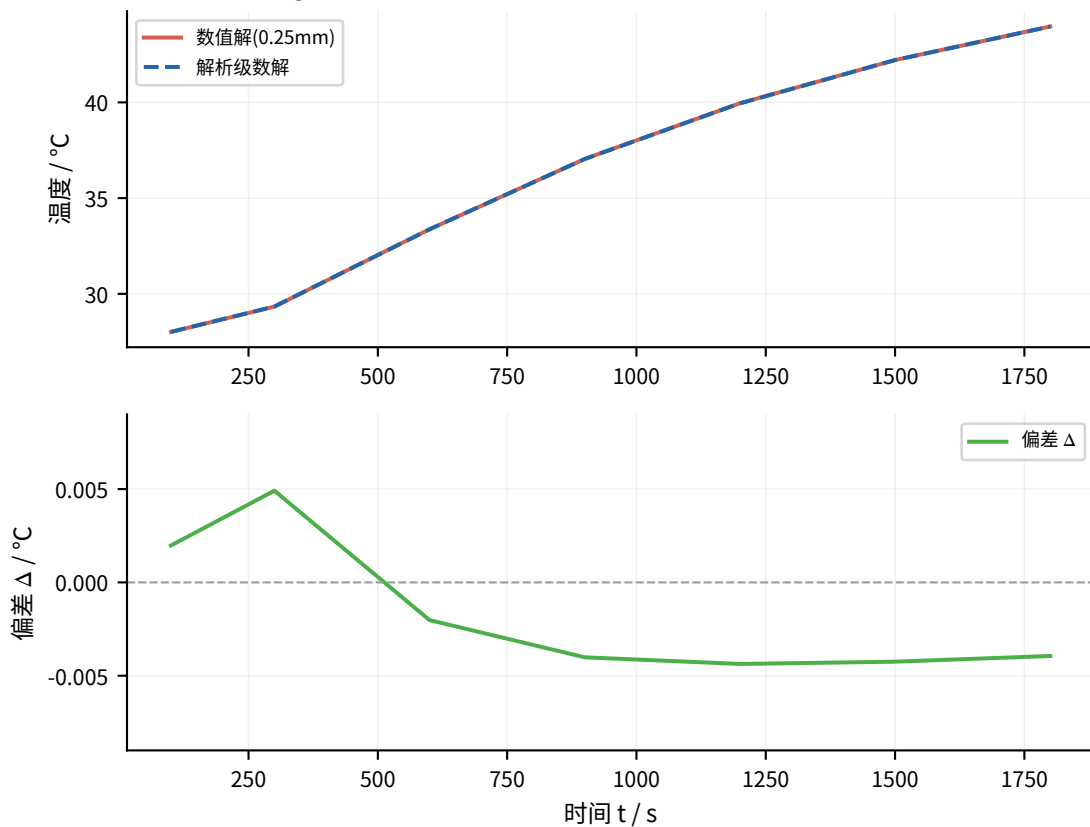
Q1 水分浓度场 $C(r,t)$ 时空分布



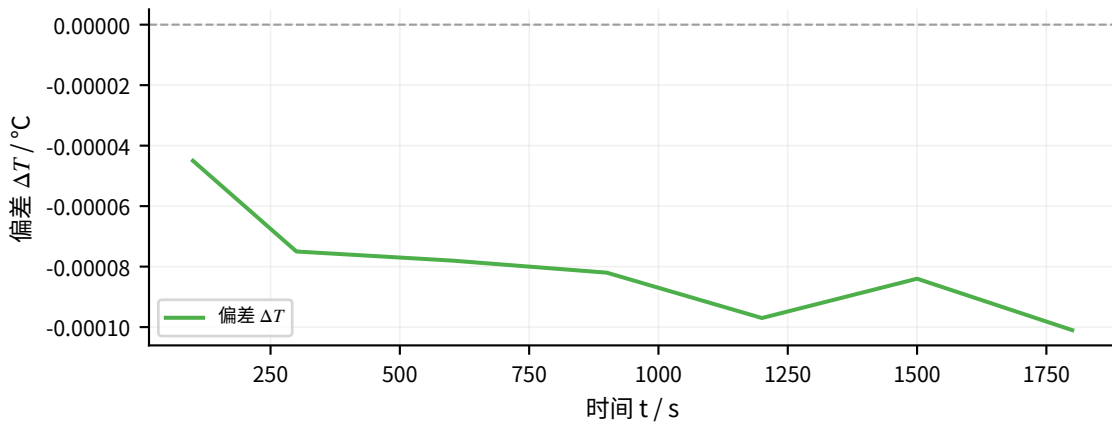
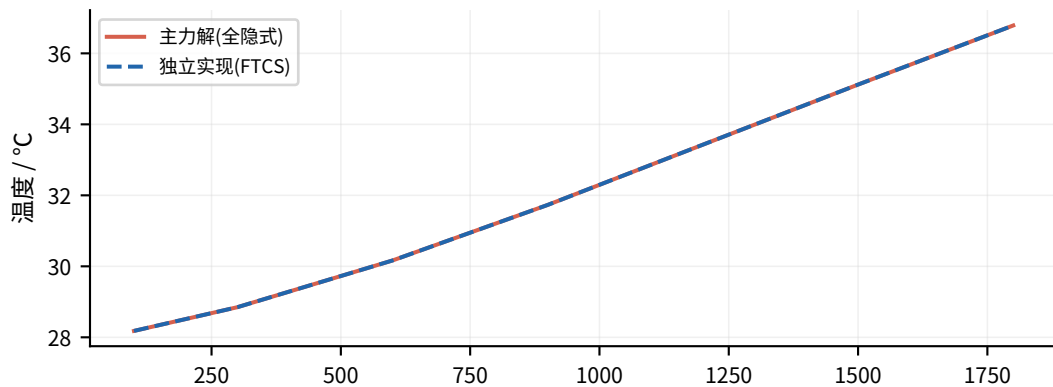
Q1 中心与表面温度、水分浓度时程曲线



Q1 温度解析对拍 (解析级数解 对比 数值解, $r=0$)

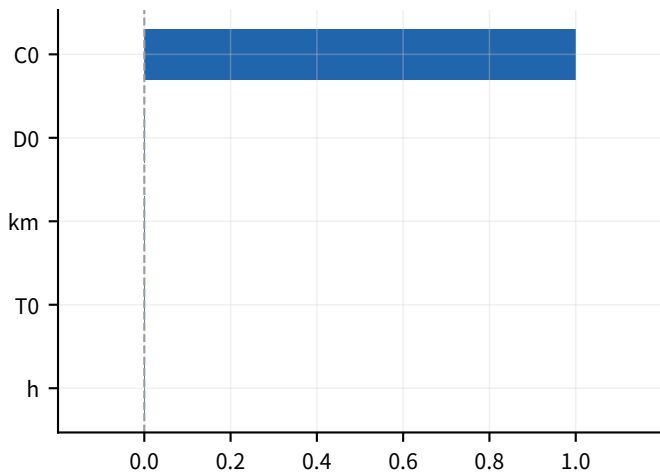


Q1 独立实现互验（表面温度， $t \leq 3h$ ）

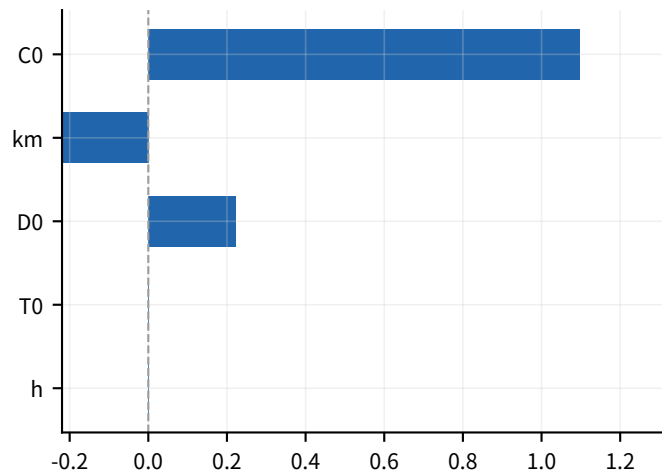


Q1 参数灵敏度龙卷风图 (OAT±20%)

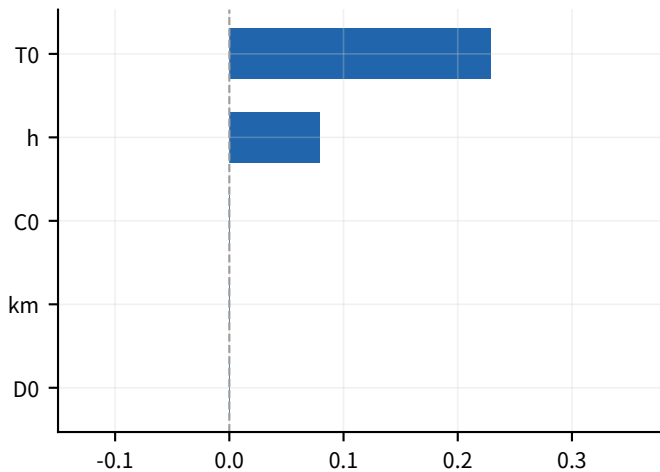
水分浓度 中心



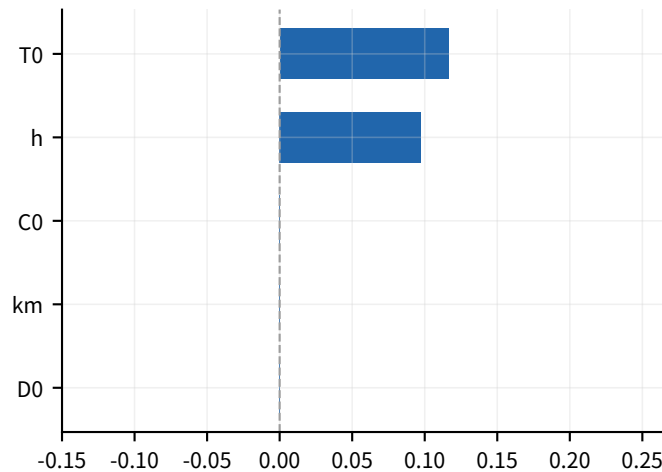
水分浓度 表面



温度 中心



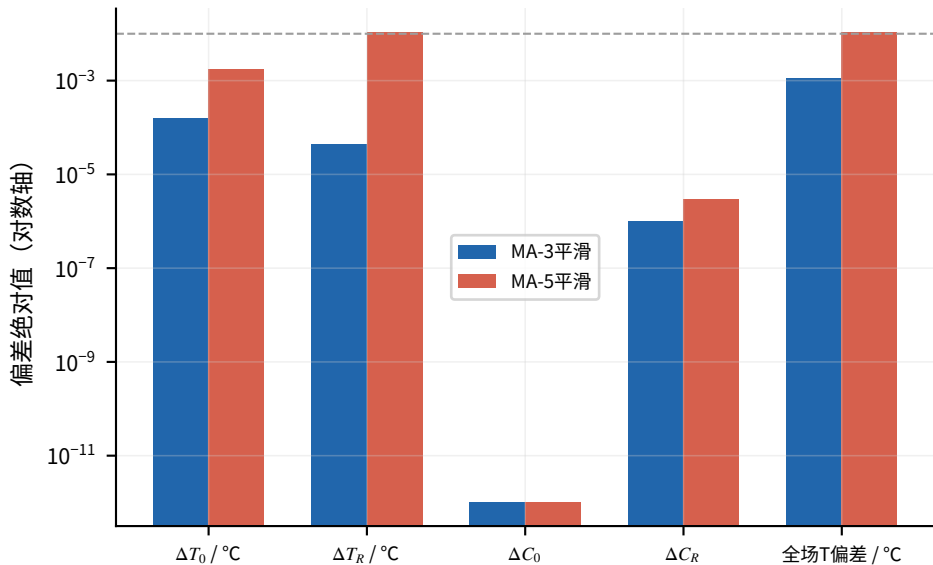
温度 表面



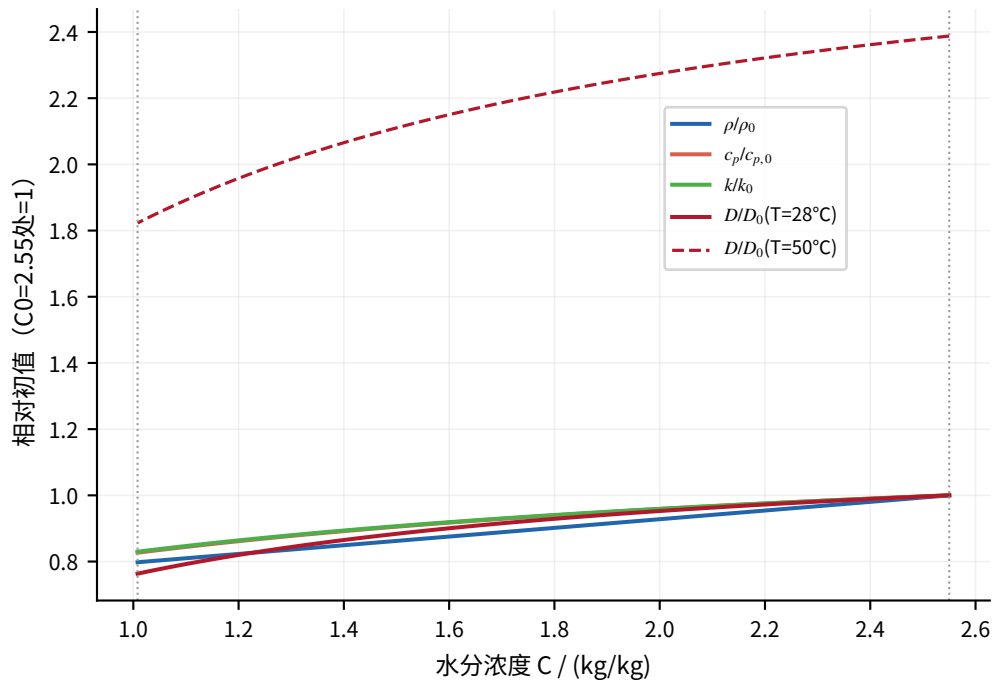
归一化灵敏度 S

归一化灵敏度 S

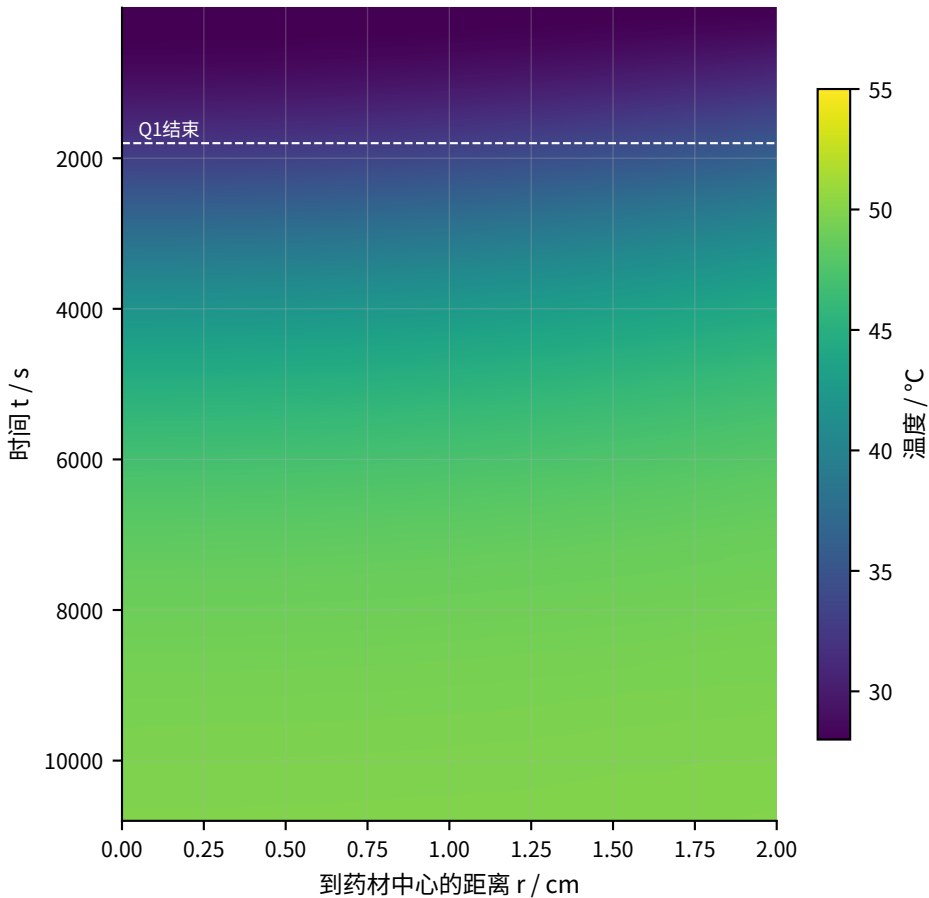
Q1 边界平滑 对比 不平滑



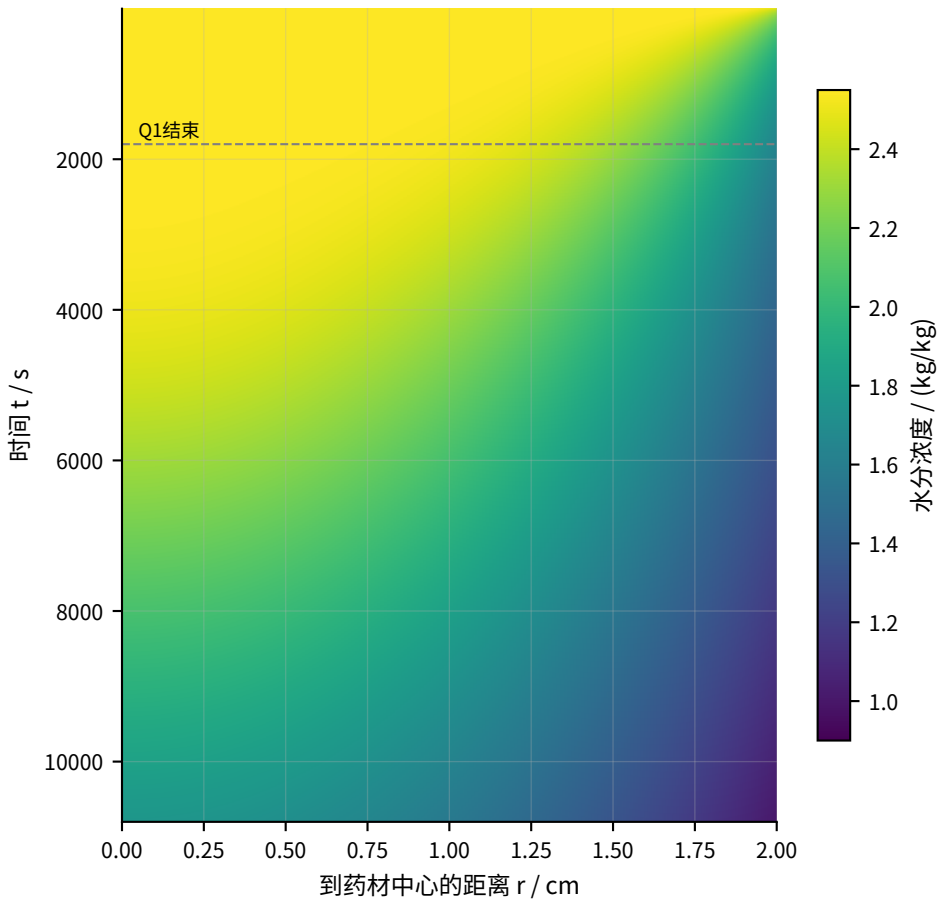
Q2 物性参数随水分浓度的演化



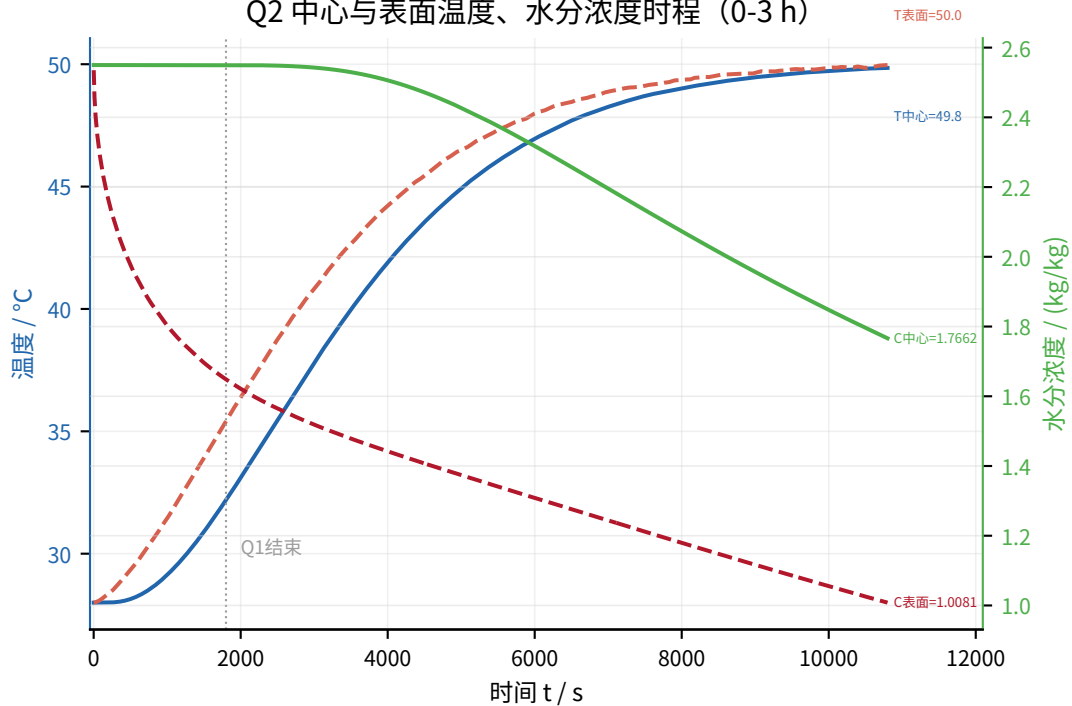
Q2 温度场 $T(r,t)$ 时空分布 (0-3 h)



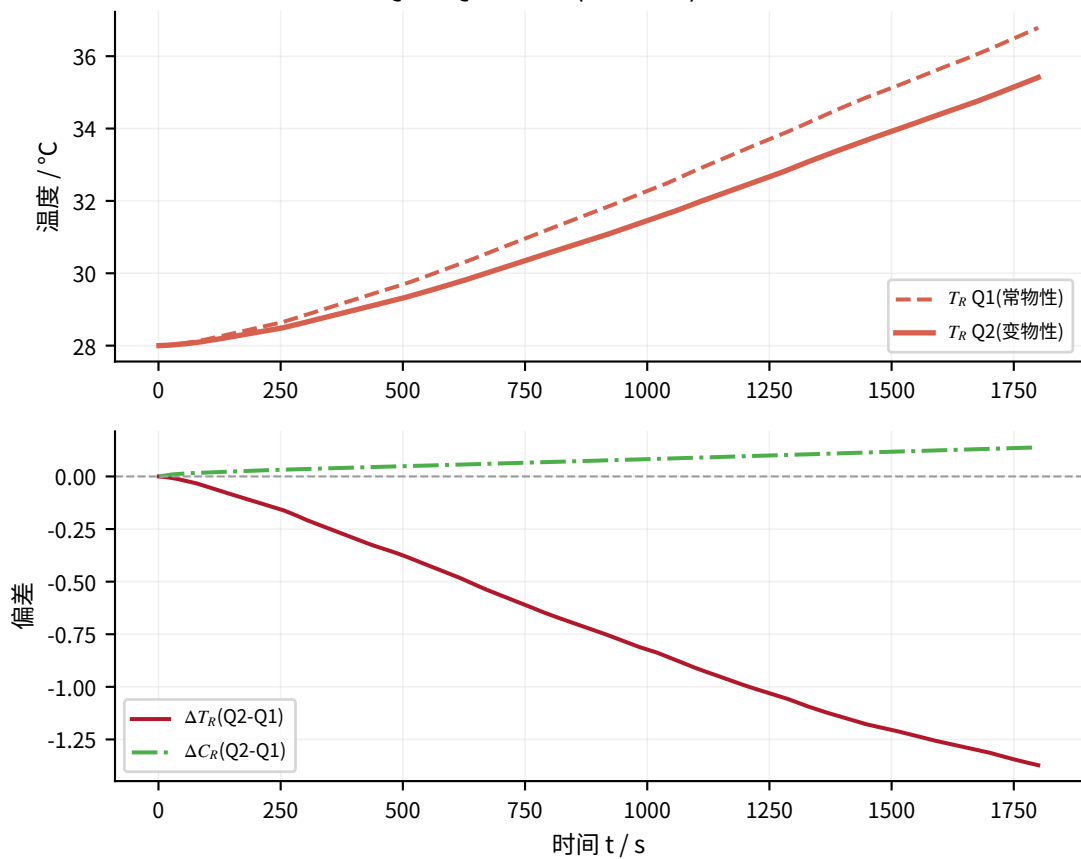
Q2 水分浓度场 $C(r,t)$ 时空分布 (0-3 h)



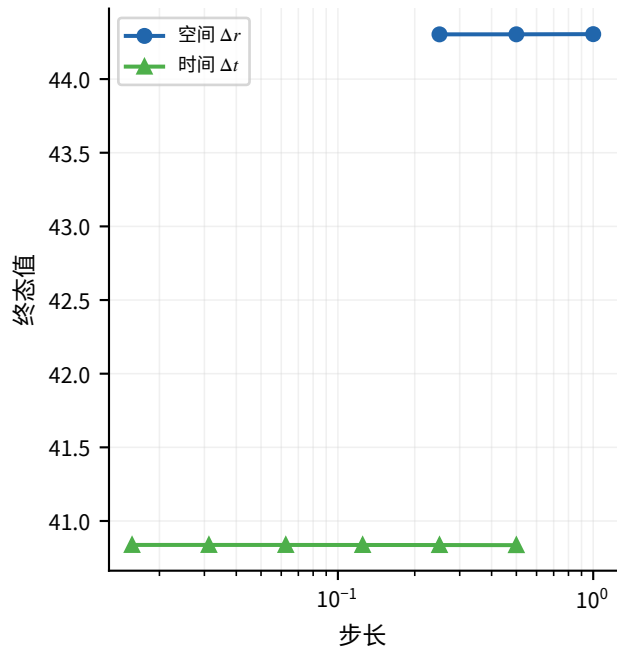
Q2 中心与表面温度、水分浓度时程 (0-3 h)



Q1与Q2重叠段(0-1800s)差异归因



(a) 收敛趋势



(b) 收敛阶

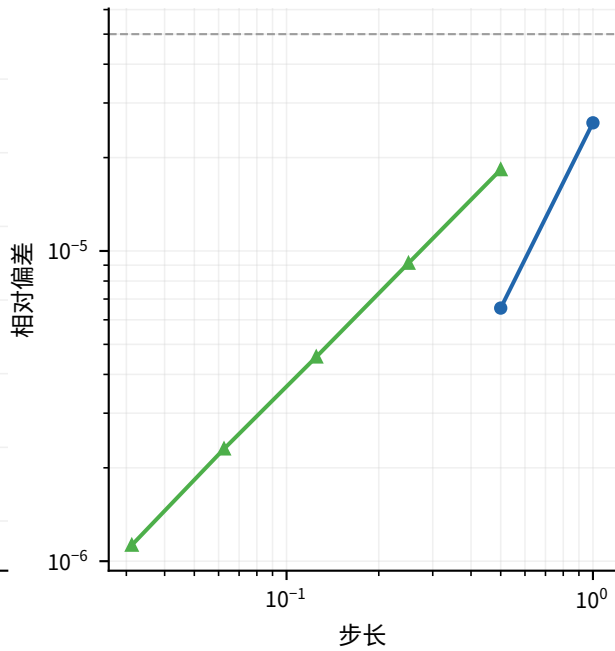
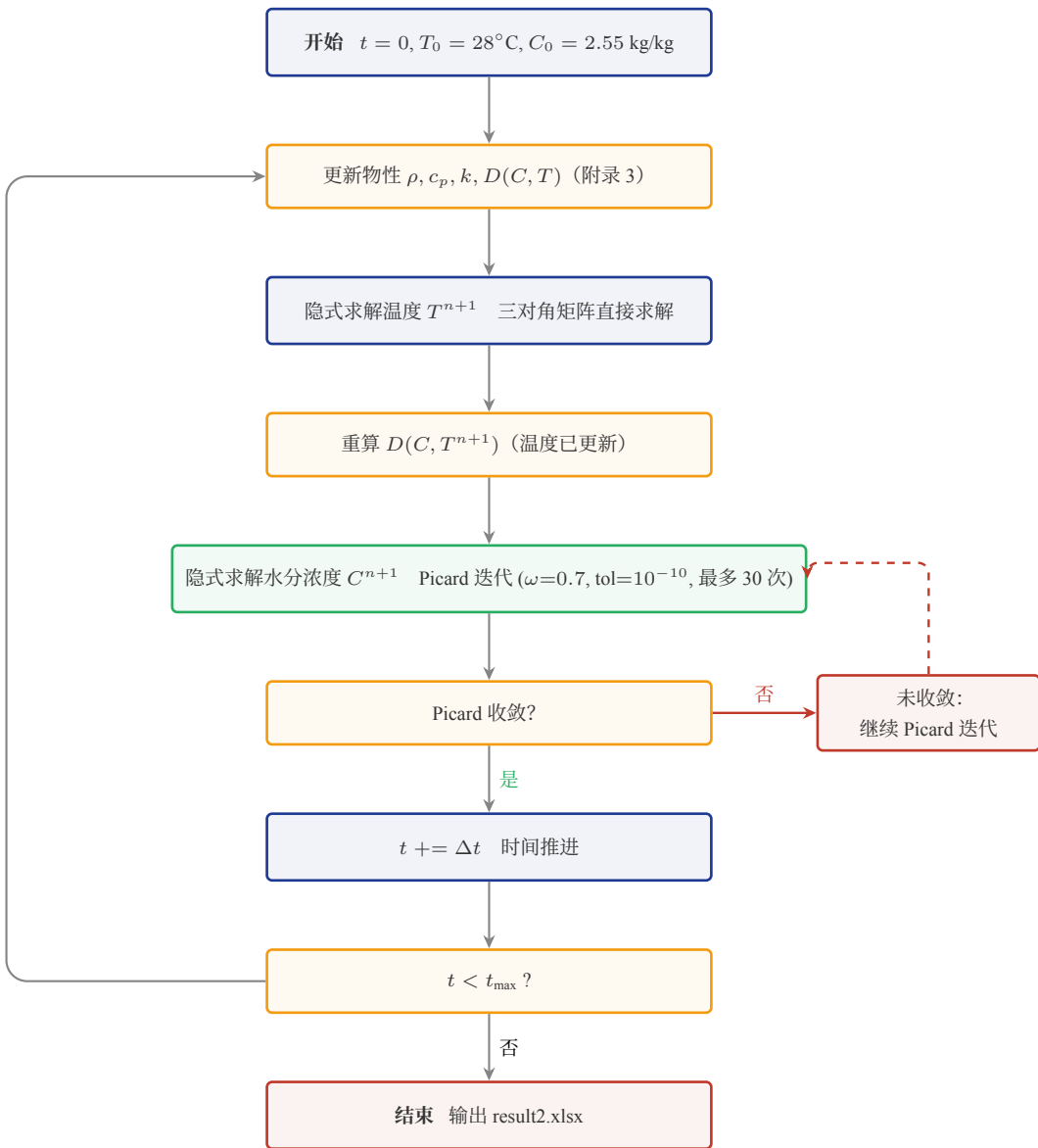
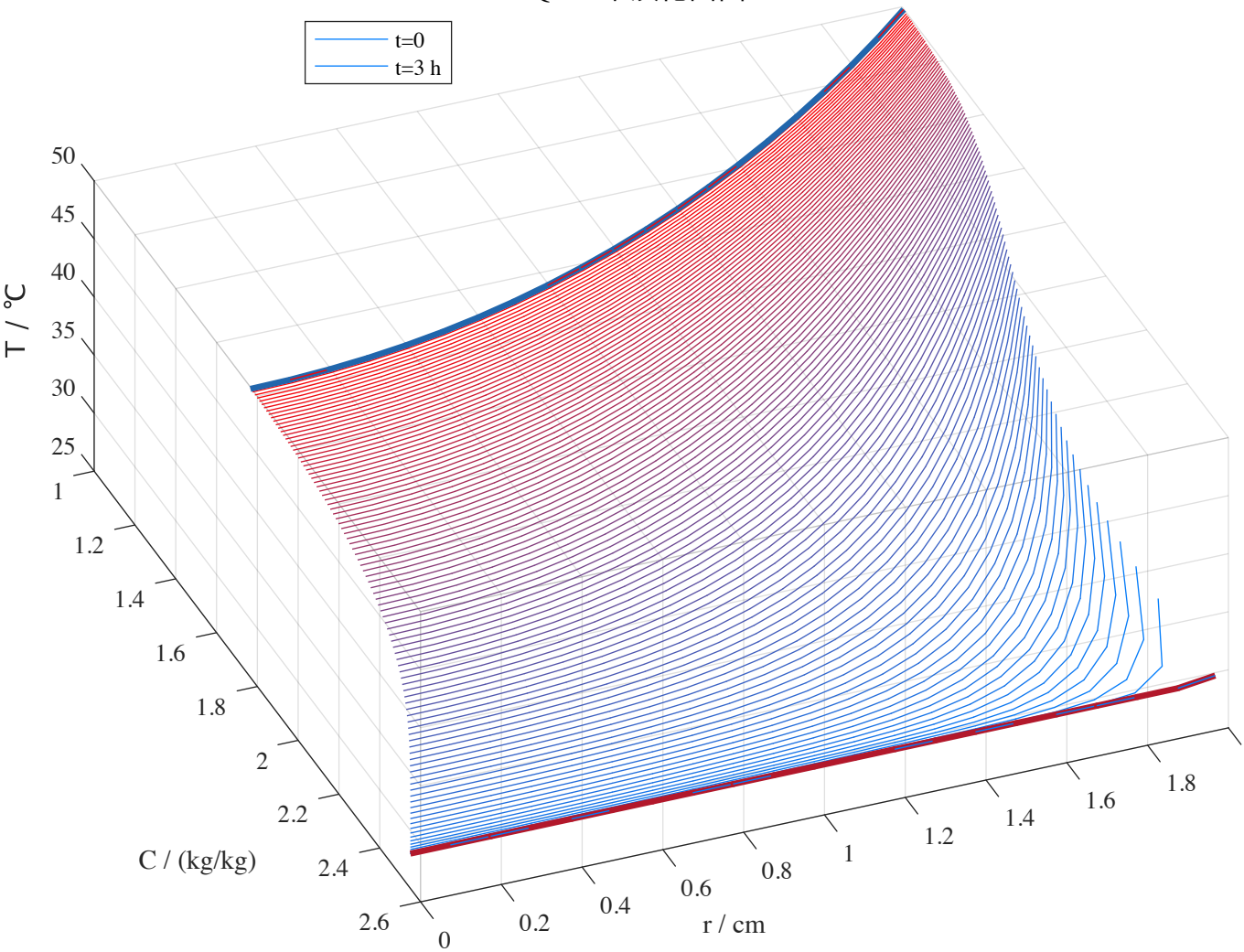


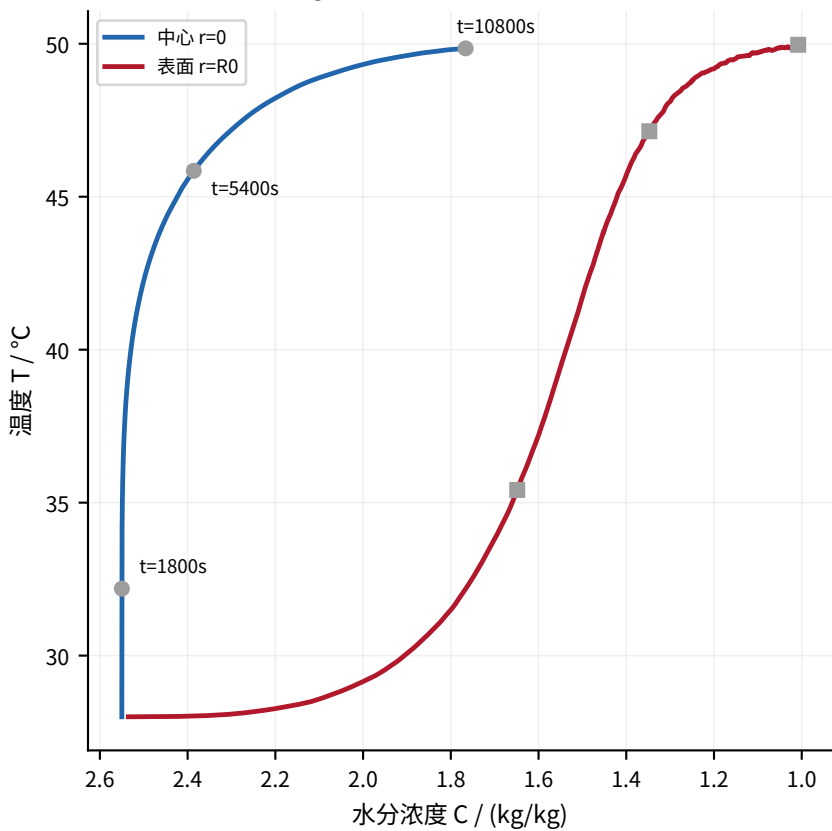
图 22: IMEX 交替推进算法流程



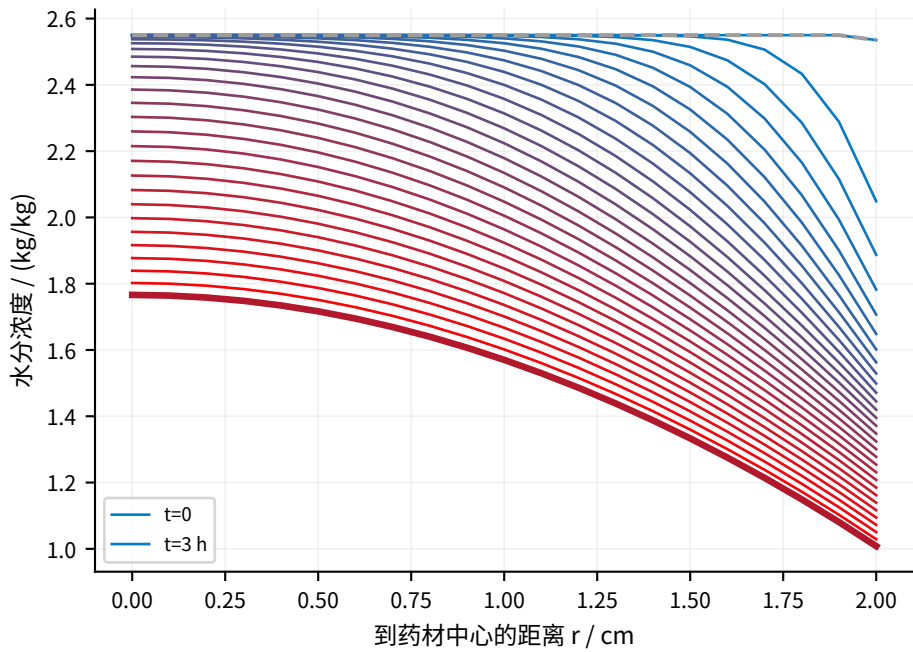
Q2 三维演化曲面 T-C-t



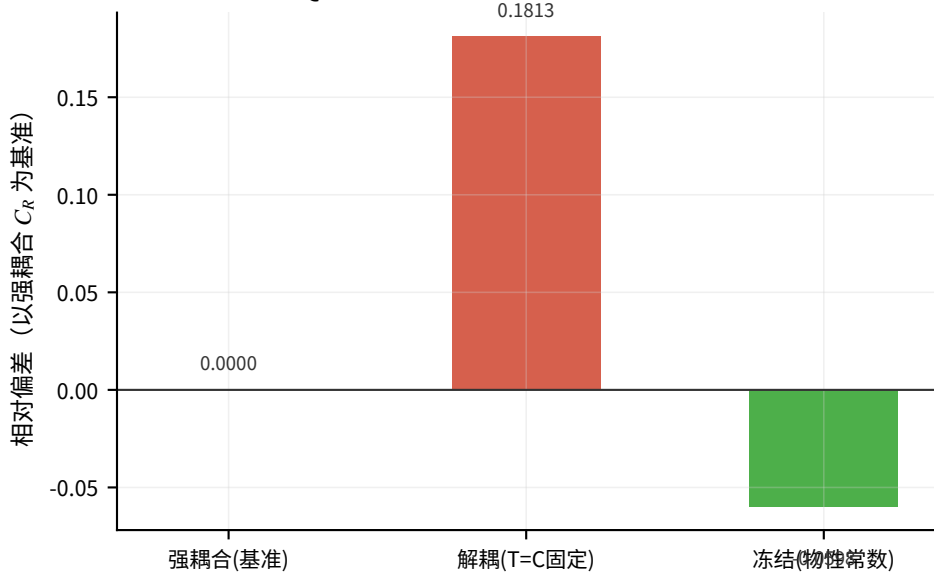
Q2 温度-水分浓度相轨迹



Q2 全断面失水演化 (浅→深=时间推进)

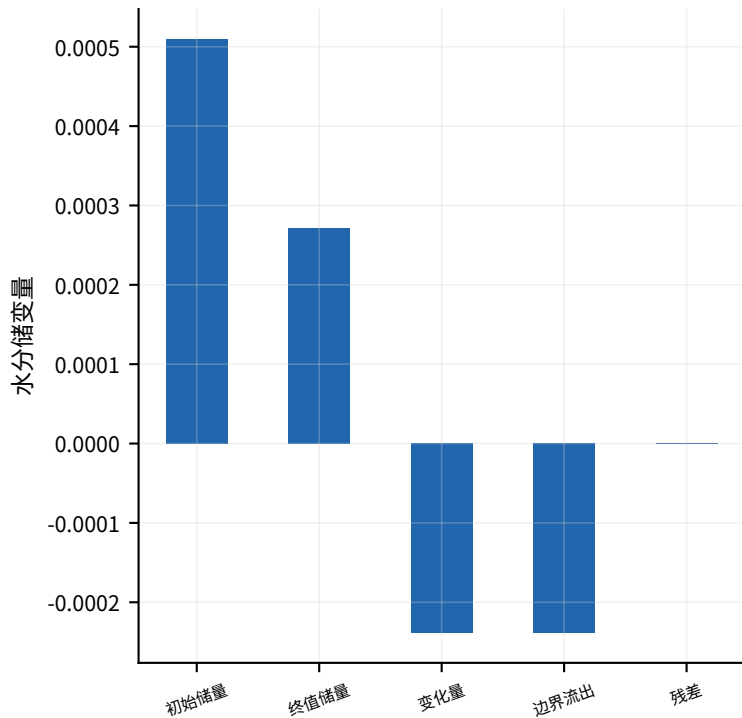


Q2 耦合强度分解 (3 h末 C_R 偏差)

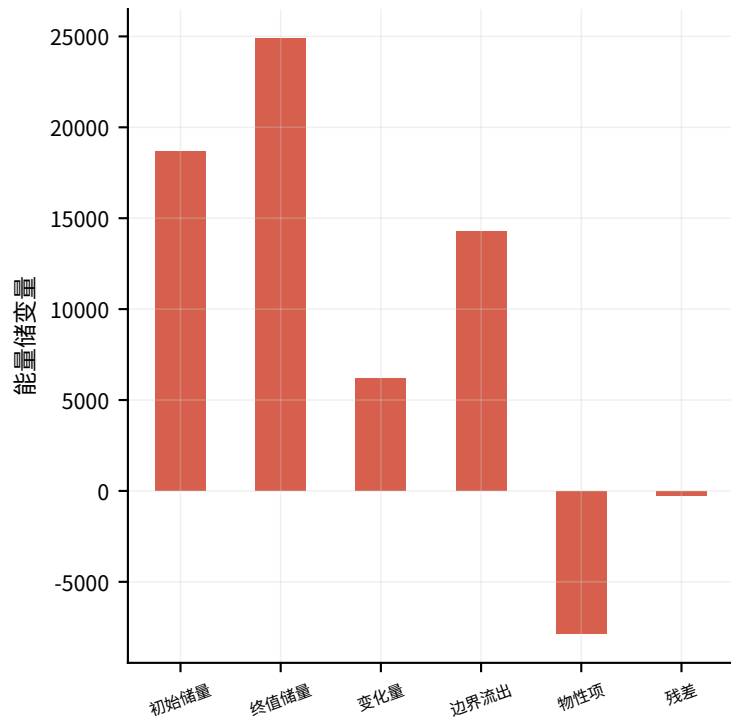


Q2 守恒性总账

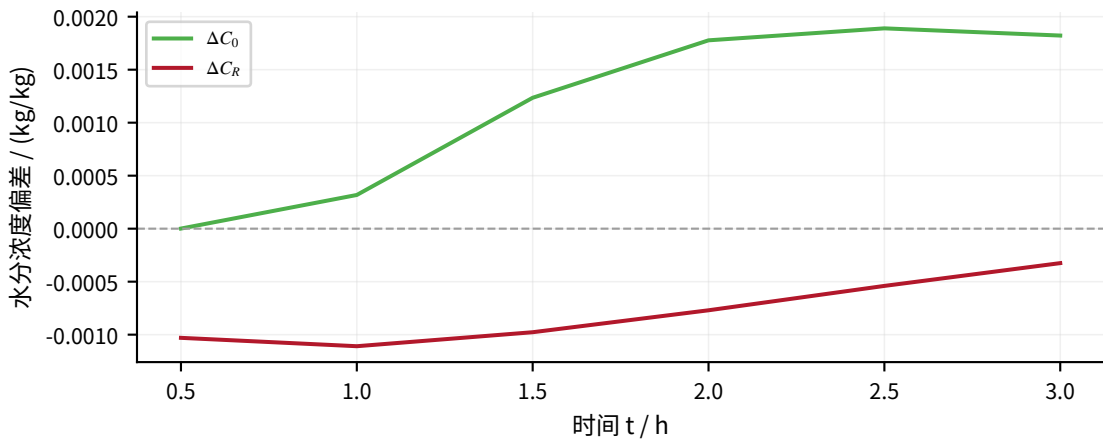
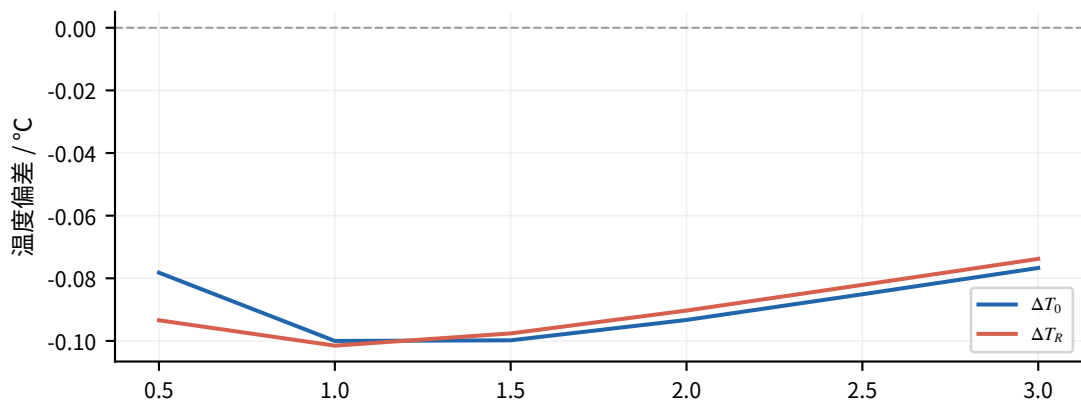
质量守恒



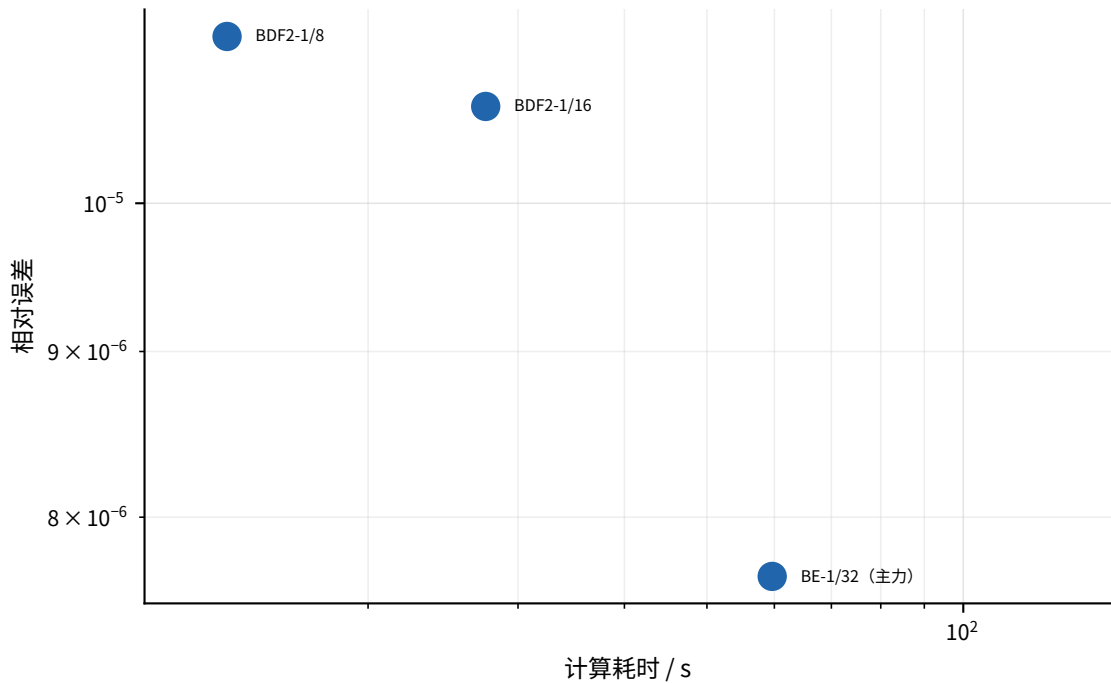
能量守恒



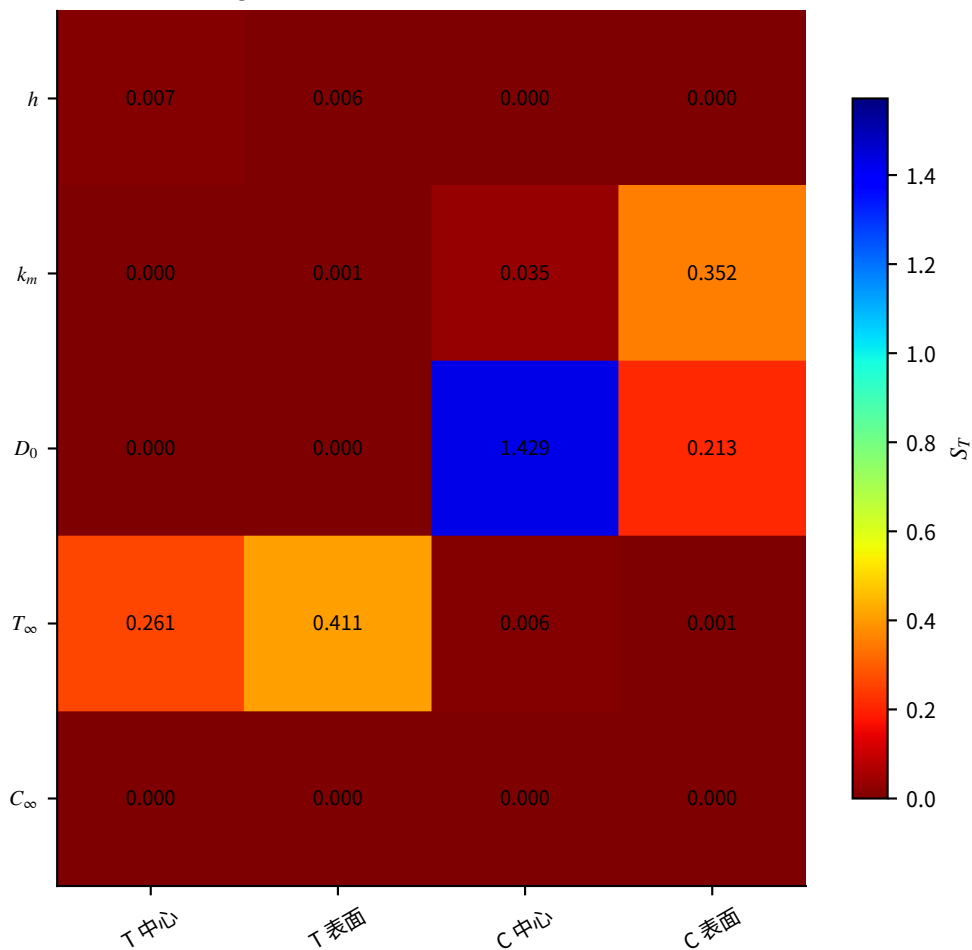
潜热效应对照（计入潜热 - 忽略潜热）



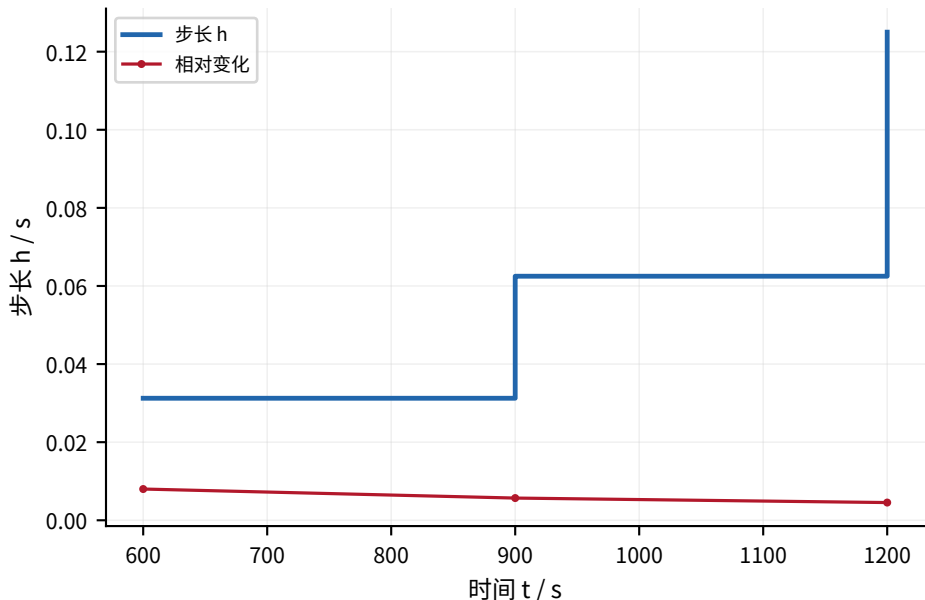
Q2 精度-效率帕累托图



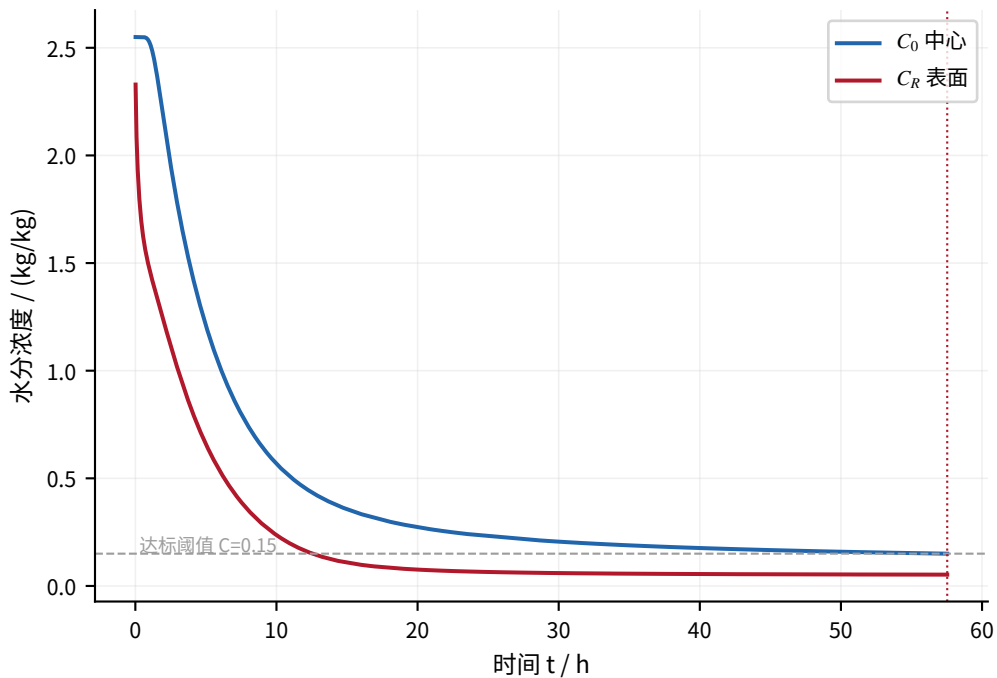
Q2 全局灵敏度 Sobol ST 热力图



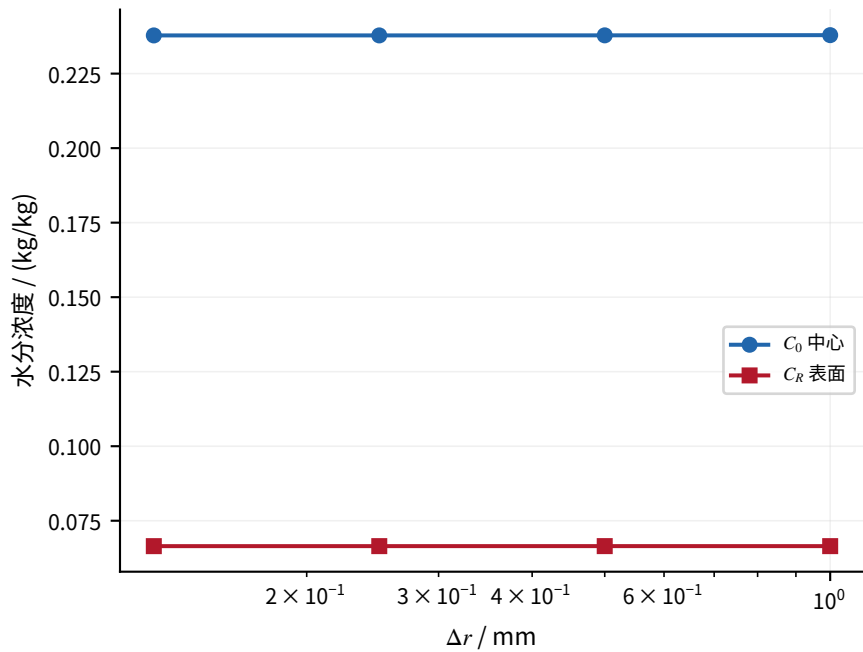
Q2 自适应步长轨迹



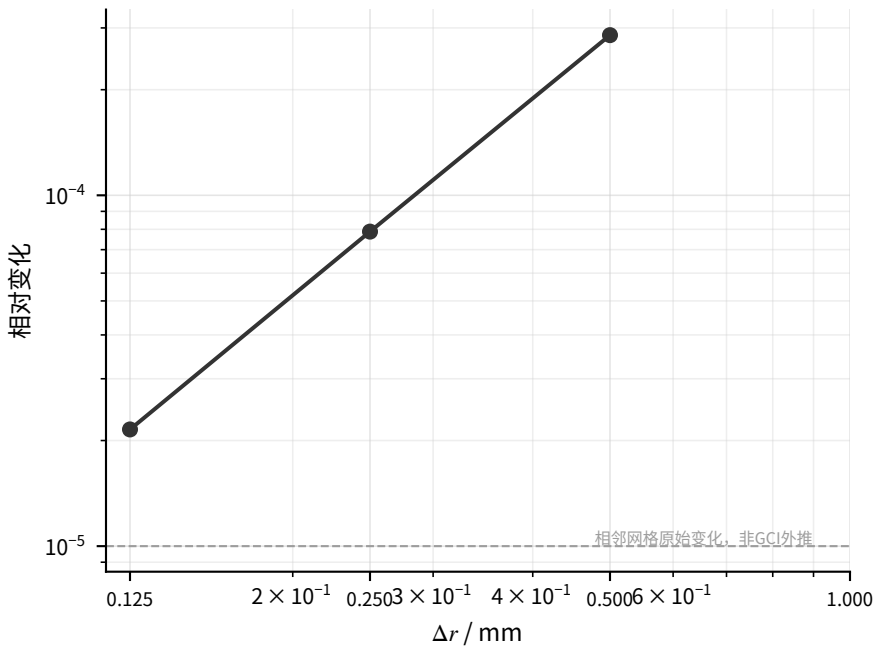
Q3 中心与表面水分浓度长时程演化



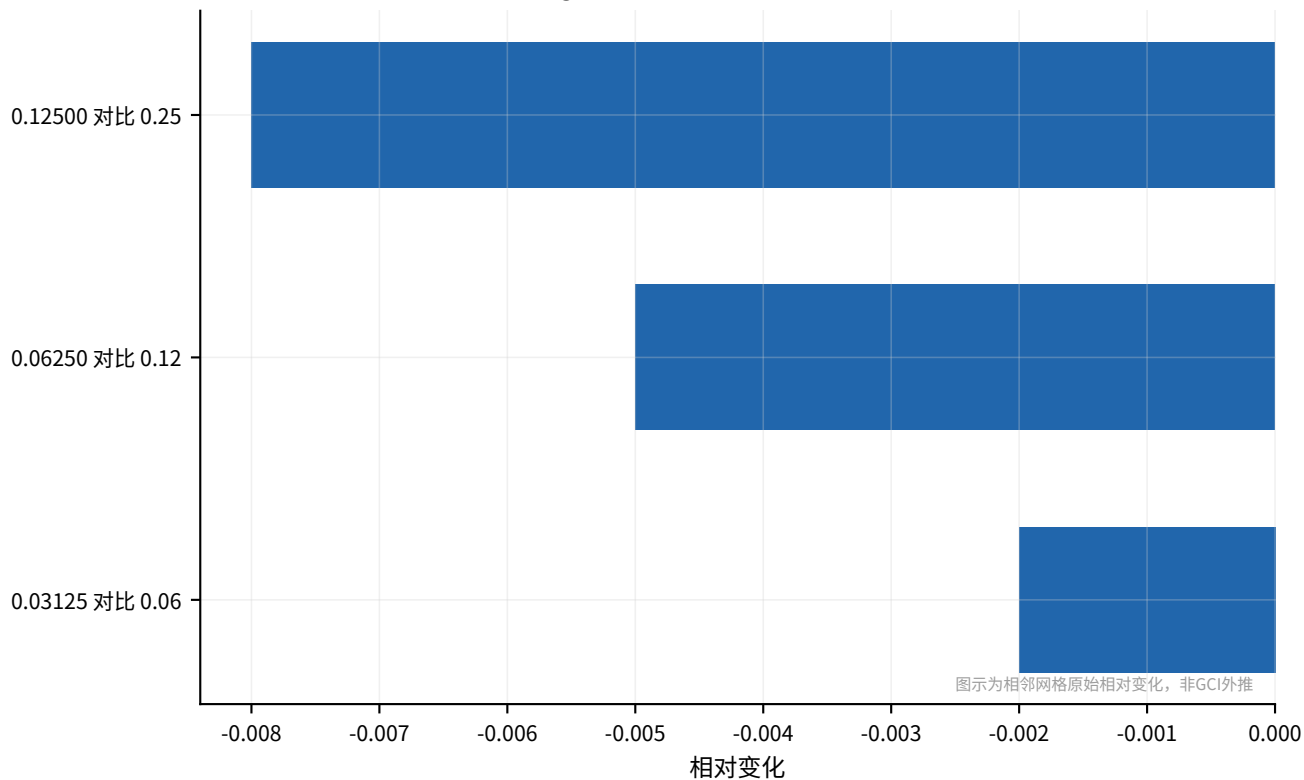
Q3 收敛性 — 固定时刻浓度随步长 (≈ 24 h)



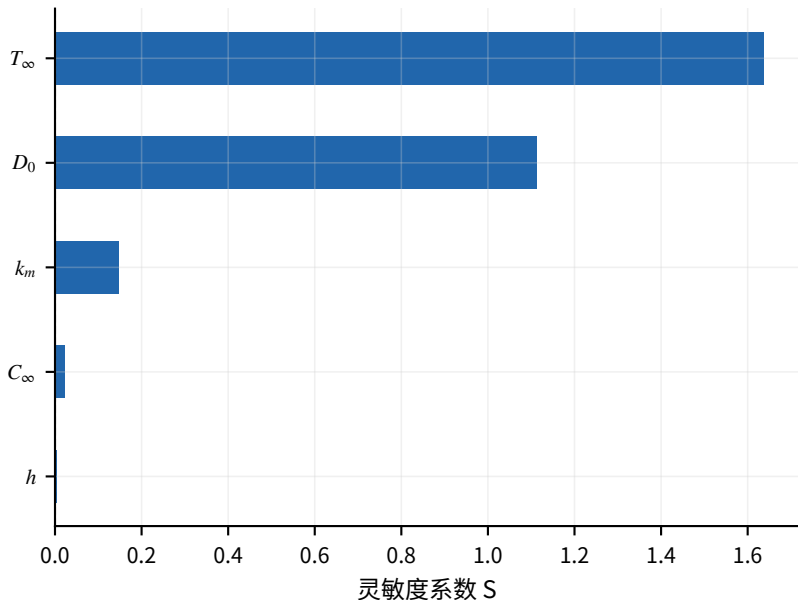
Q3 收敛性 — 收敛阶



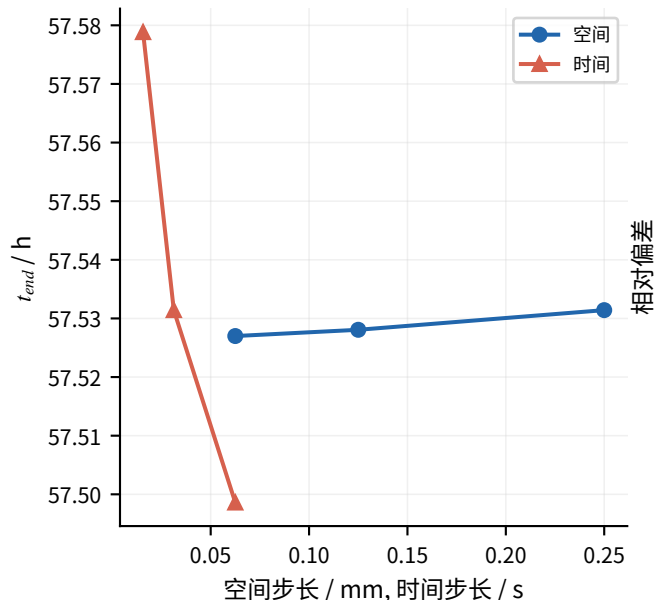
Q3 收敛性 — 各位置收敛情况



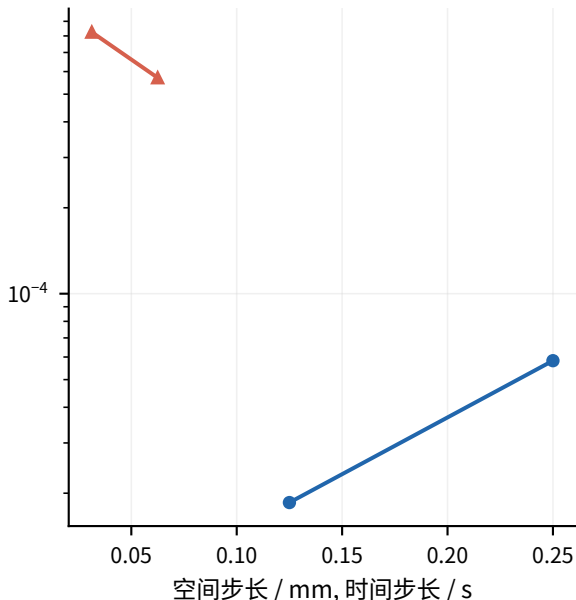
Q3 参数灵敏度



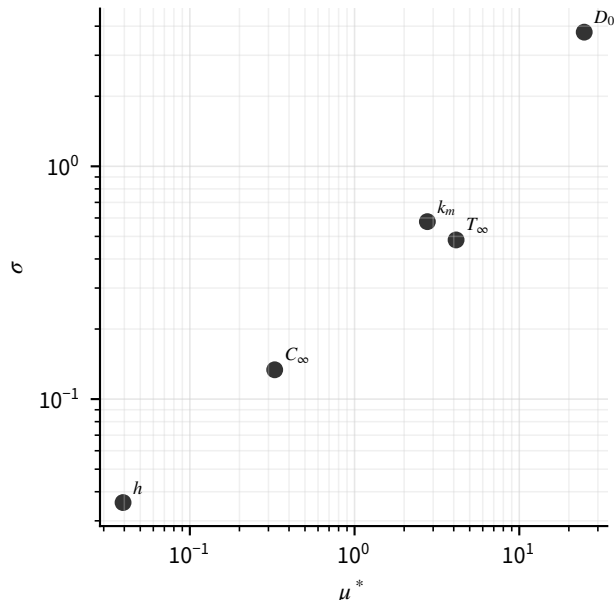
(a) 终态值



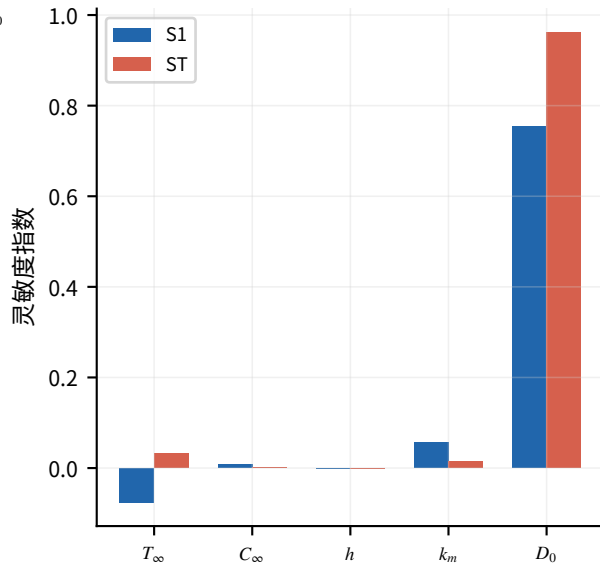
(b) 收敛趋势



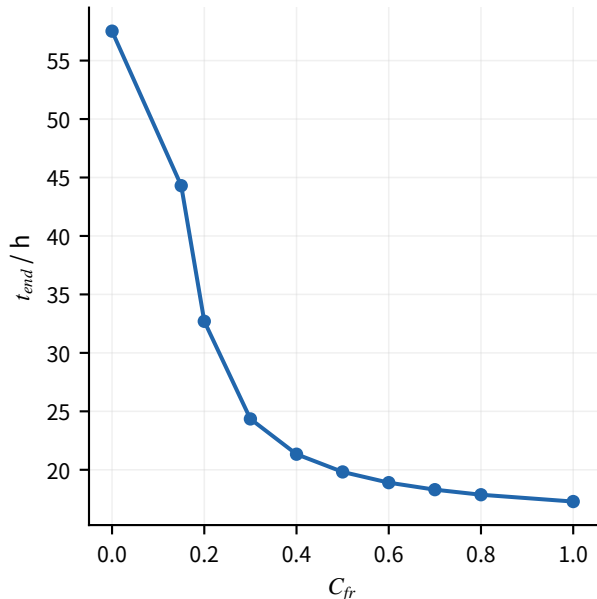
Morris $\mu^*-\sigma$



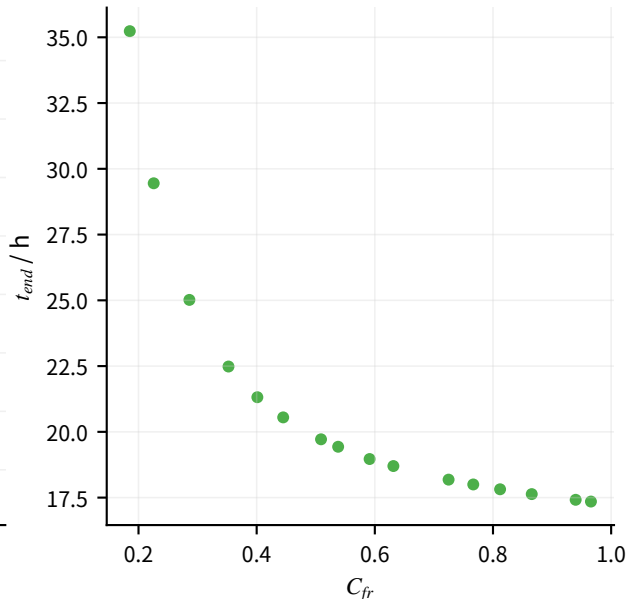
Sobol S1 对比 ST



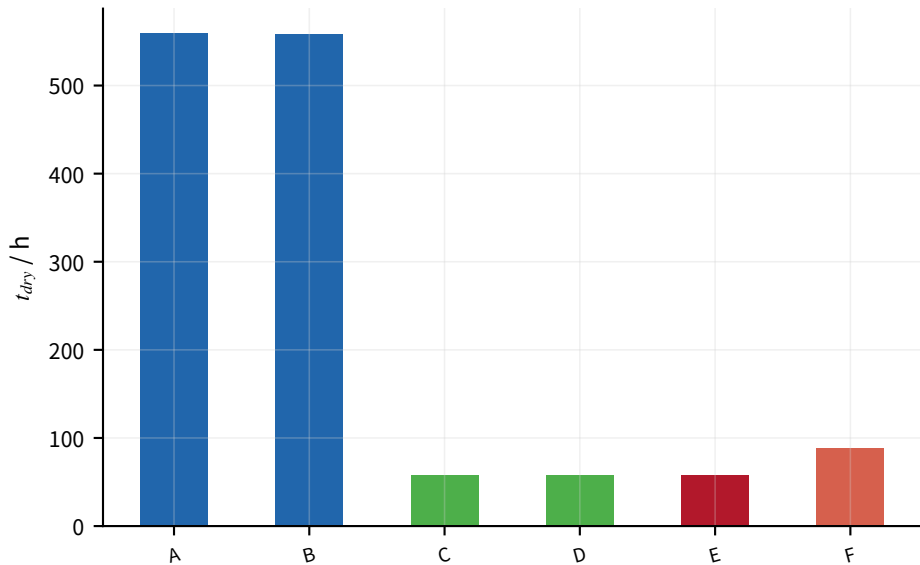
(a) 截断族曲线



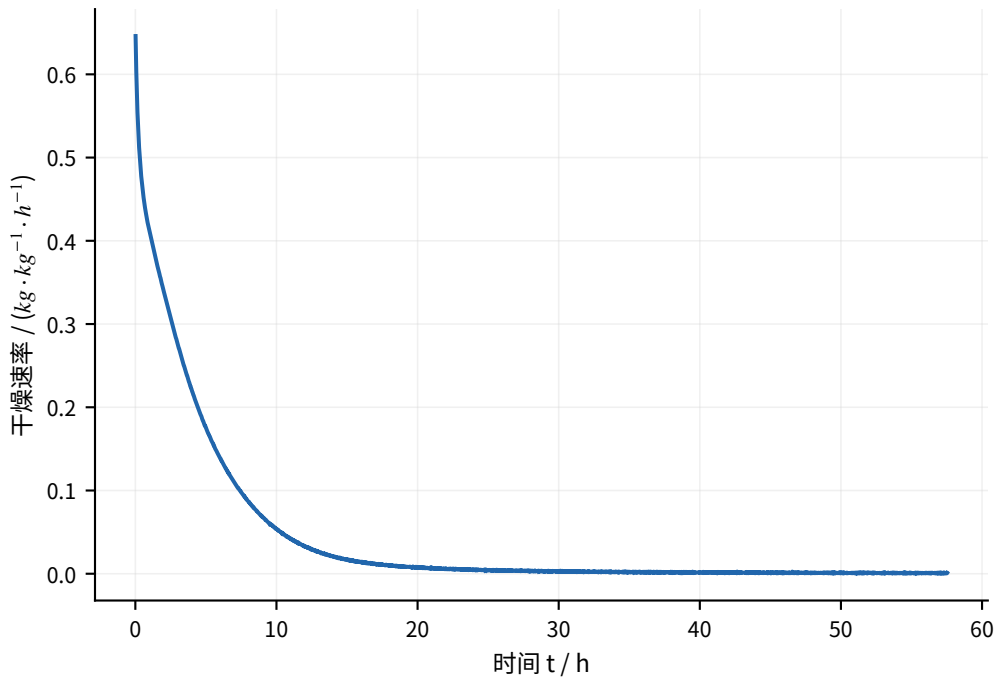
(b) LHS (n=16)



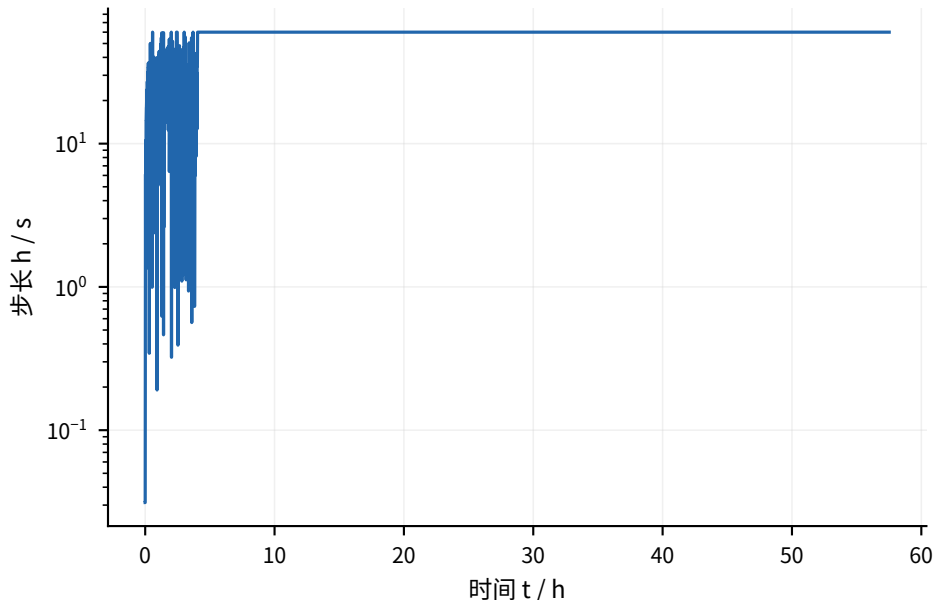
Q3 界面取法×网格分辨率对照



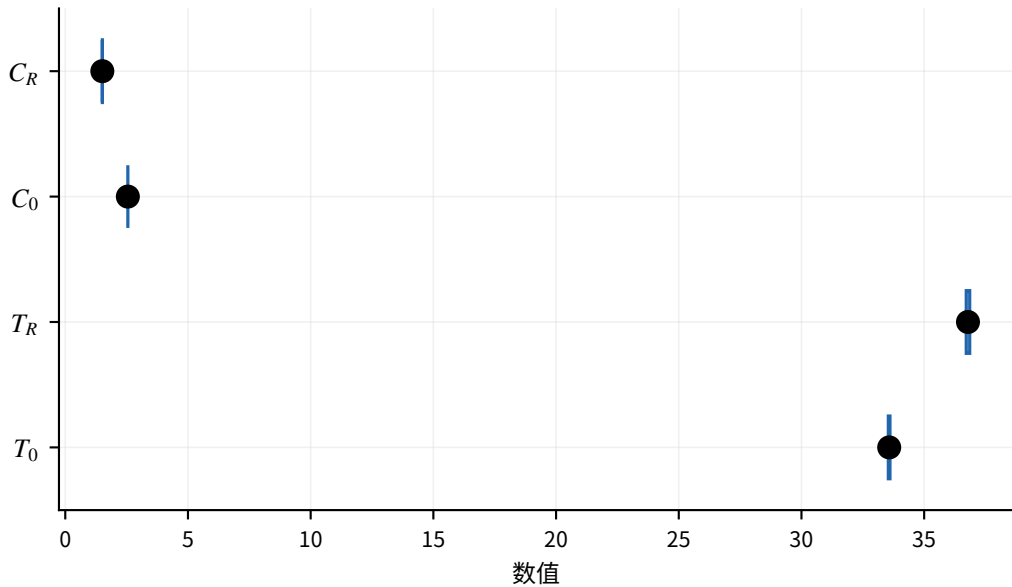
Q3 干燥速率曲线



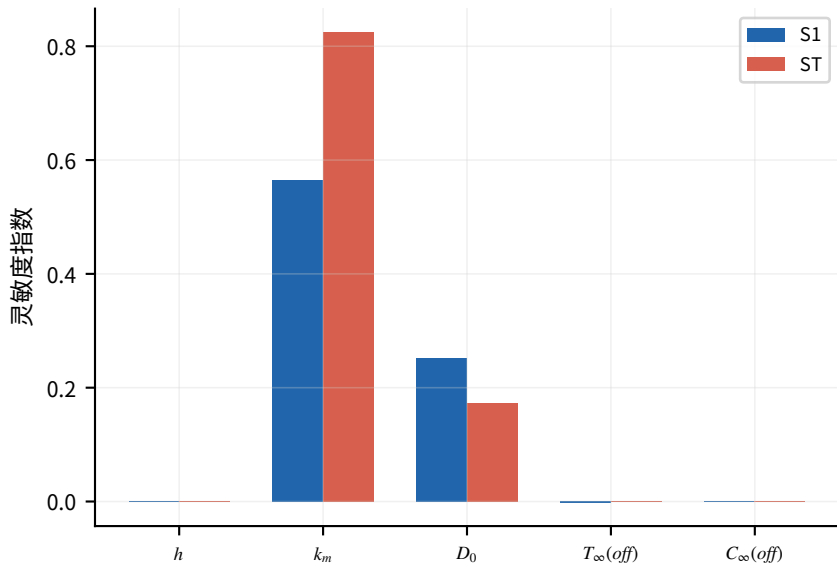
Q3 误差驱动自适应步长轨迹



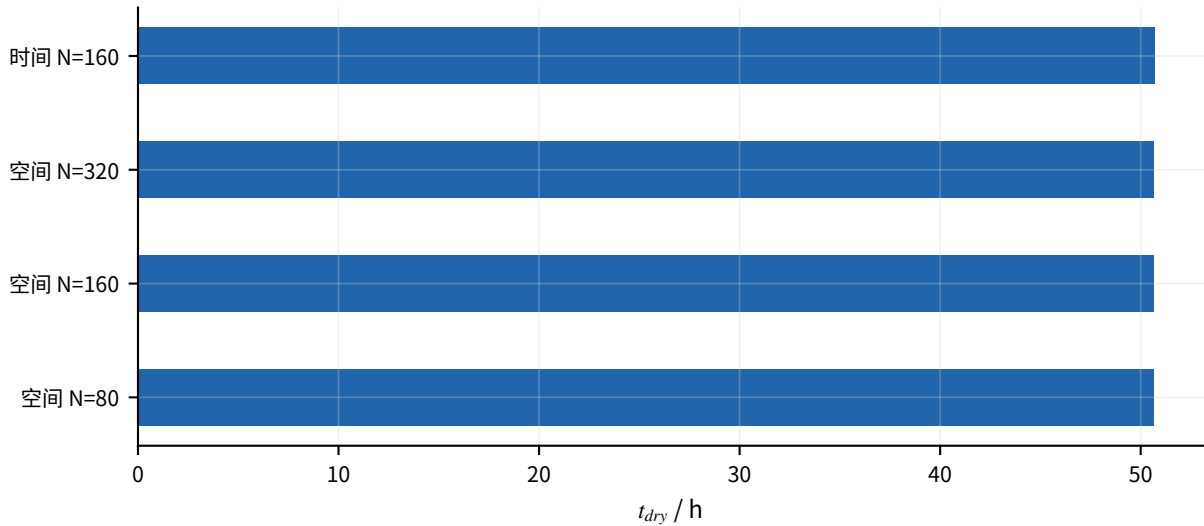
Q1 边界不确定性传播区间



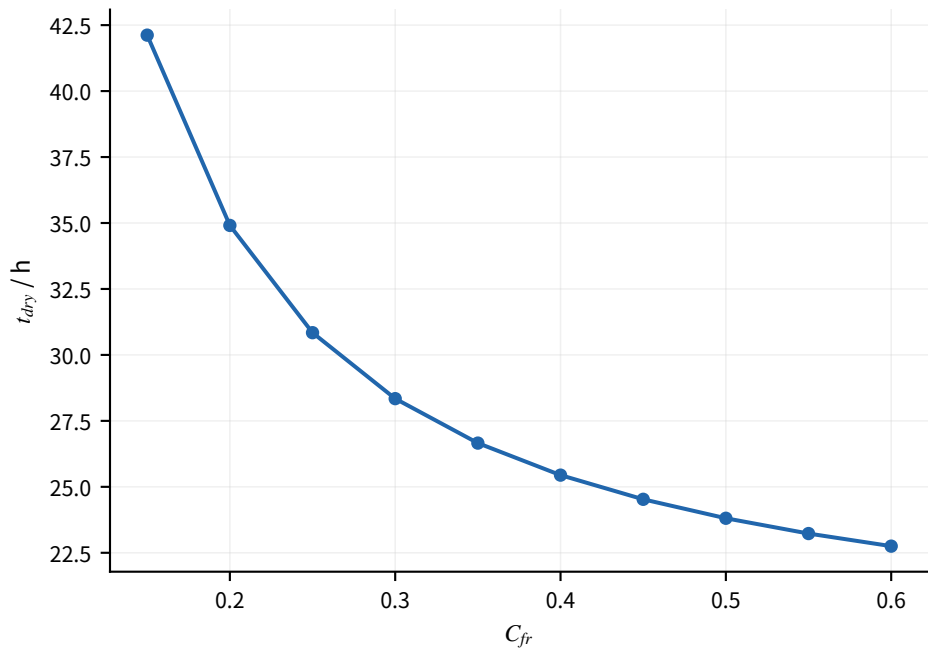
Q1 全局灵敏度 Sobol (C_R)



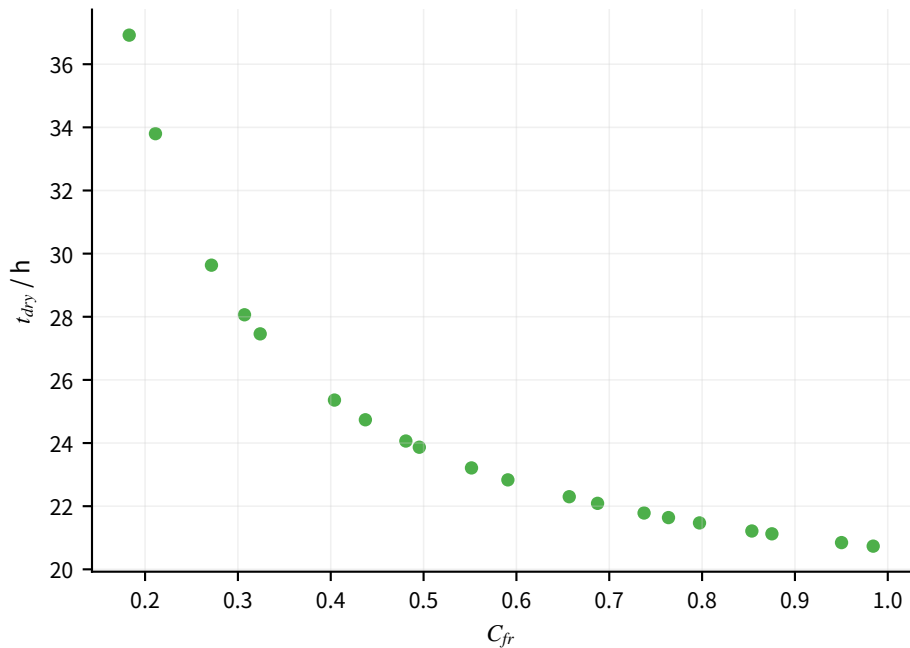
Q4 离散误差带 (GCI)



Q4 低C端外推不确定度 — 截断族曲线

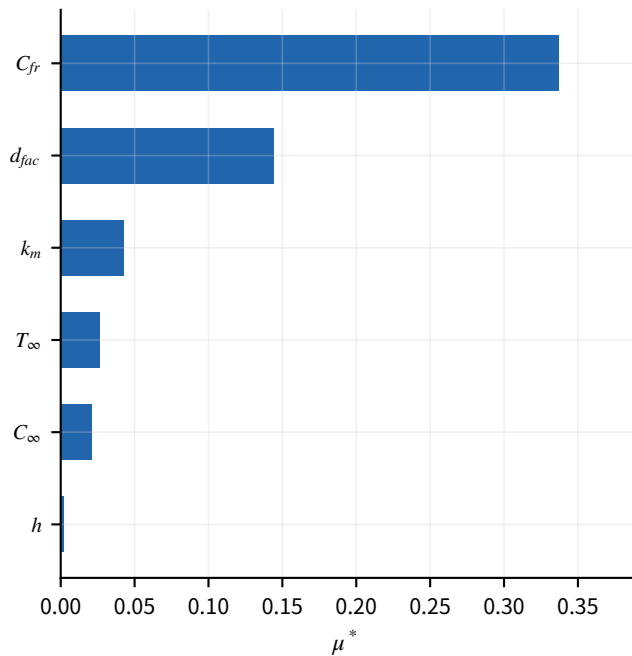


Q4 低C端外推不确定度 — LHS分位 (n=20)

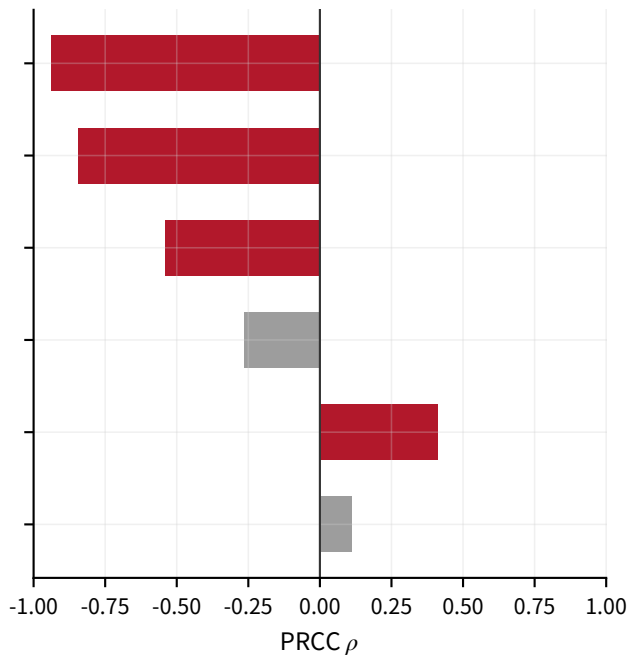


Q4 全局灵敏度 (Morris + PRCC + Sobol)

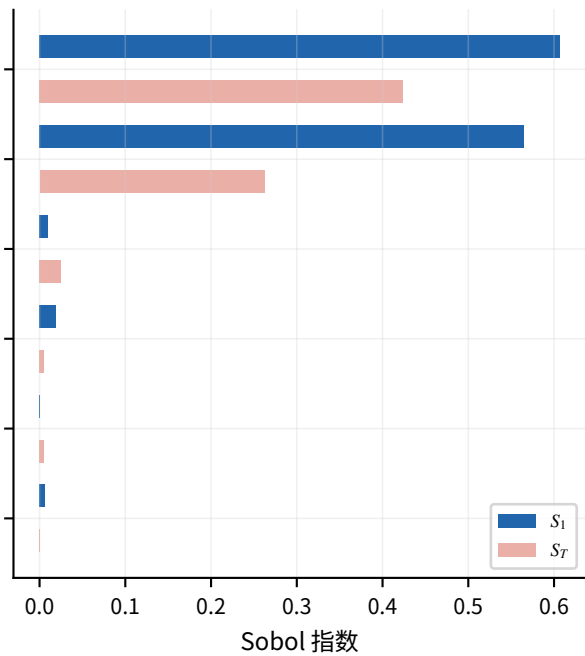
(a) Morris μ^*



(b) LHS-PRCC



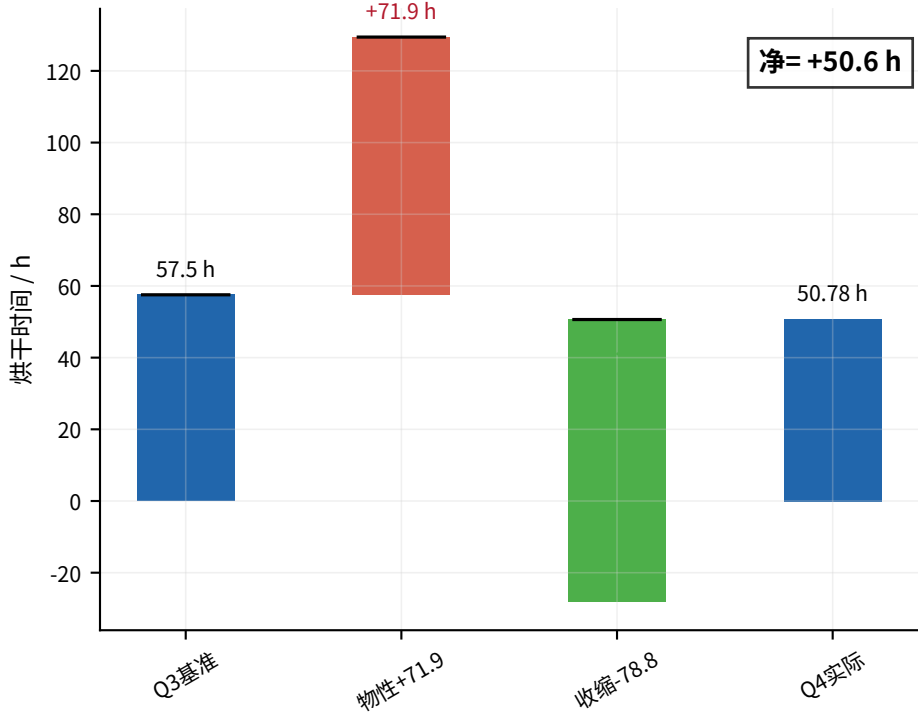
(c) Sobol S_1 / S_T



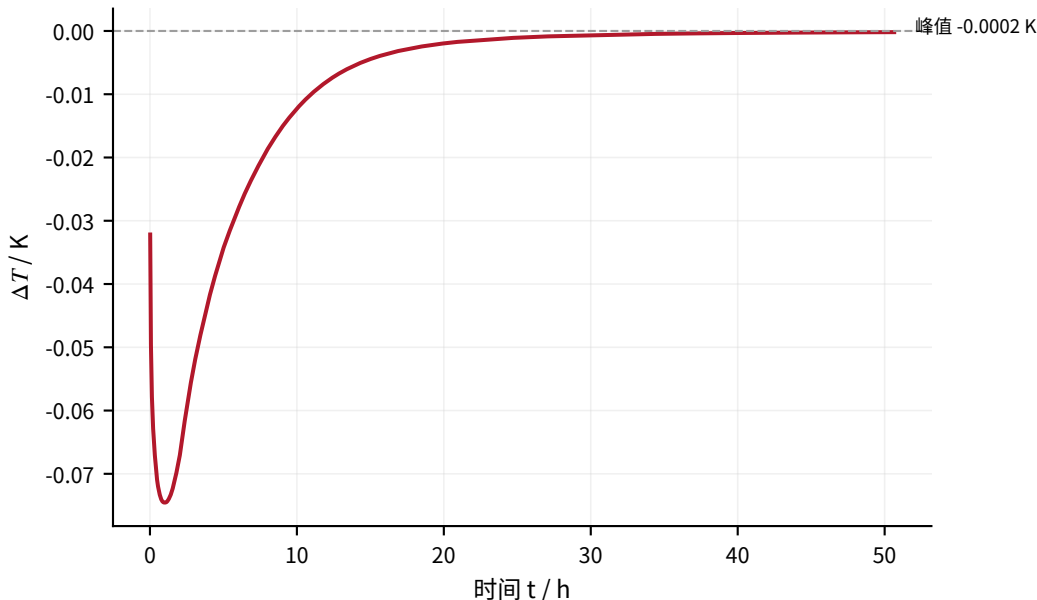
Q4 收缩-物性双效应分解

+71.9 h

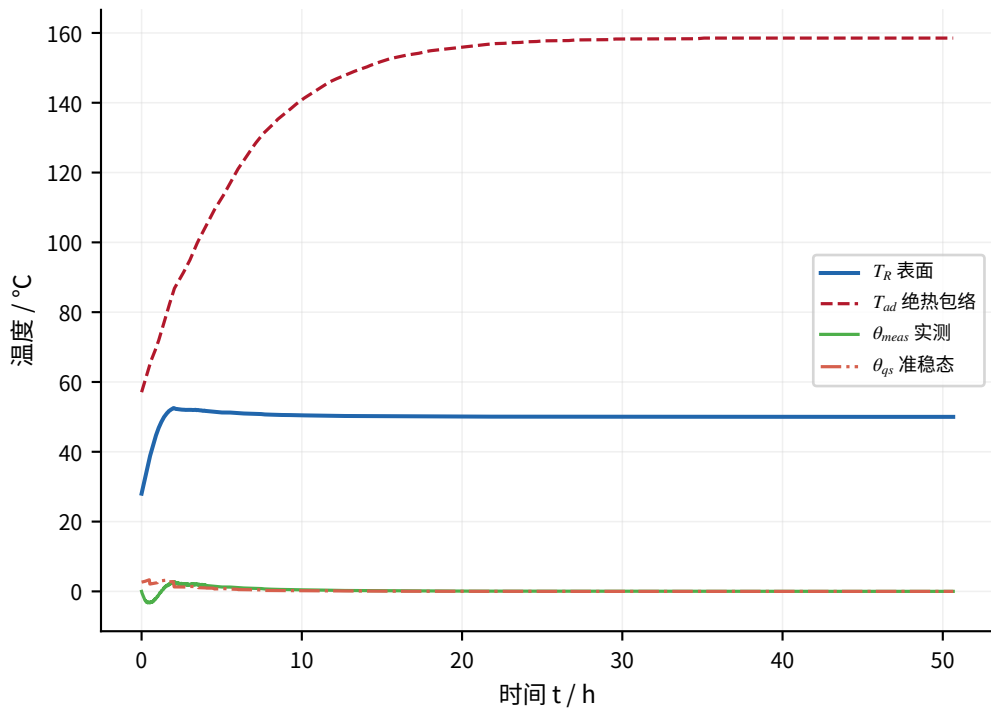
净= +50.6 h



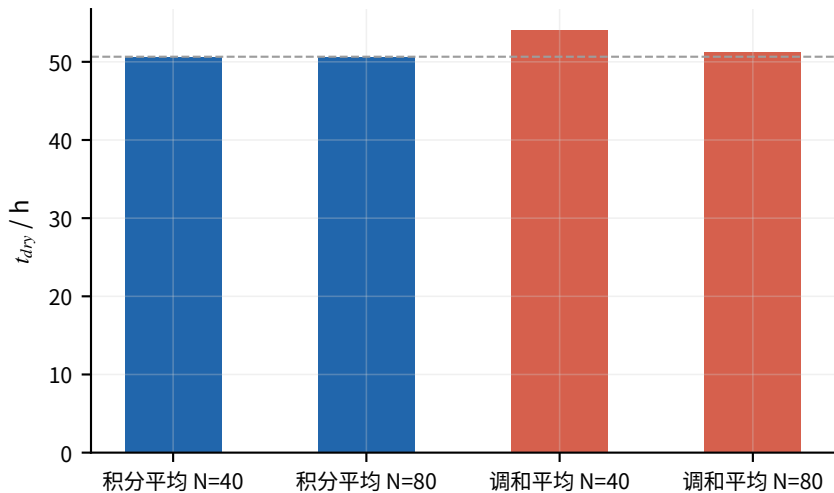
Q4 潜热中段温降曲线



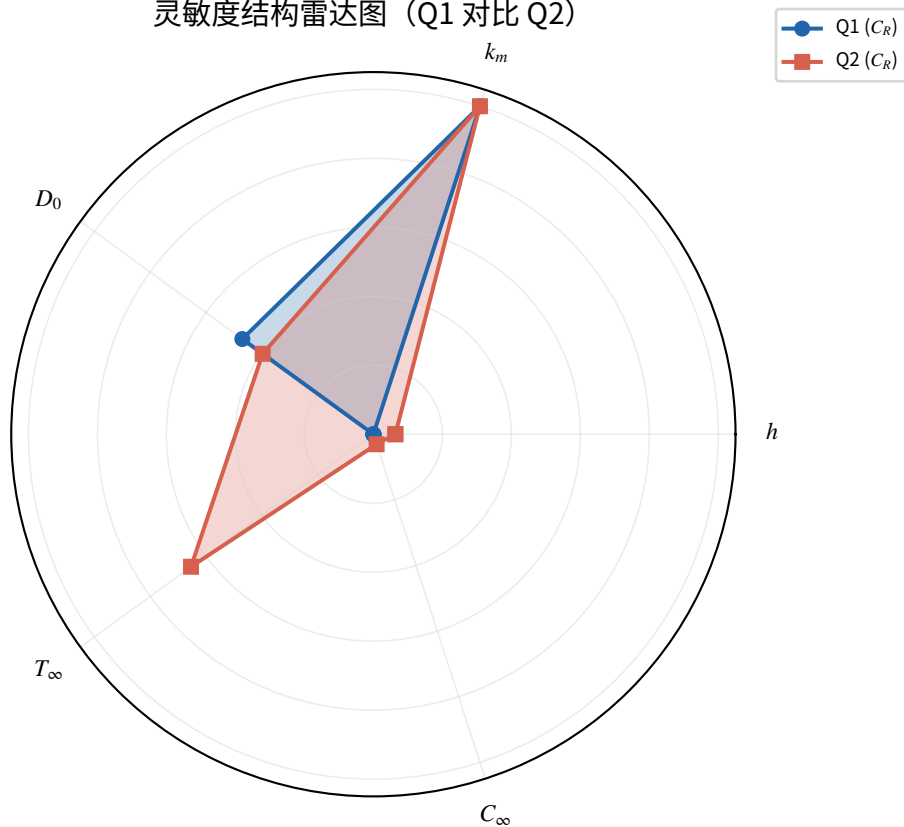
Q4 压缩温升机理与绝热包络



Q4 界面取法 × 网格对照



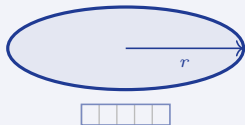
灵敏度结构雷达图 (Q1 对比 Q2)



各层按自身最大 $|S|$ 归一；仅比较形状与排序

物理域 r (随时间收缩)

$$t_0 \quad R(t_0) \approx 1.8 \text{ cm}$$



$$t_1 \quad R(t_1) \approx 1.4 \text{ cm}$$



$$t_{\text{dry}} \quad R(t_{\text{dry}}) \approx 1.08 \text{ cm}$$



网格随收缩同步加密

坐标变换

$$\xi = \frac{r}{R(t)}$$

计算域 $\xi = r/R(t)$ (固定)



$\xi \in [0, 1]$, 不随时间变化
 $\Delta\xi = 1/160$, 节点数不变



$\xi \in [0, 1]$, 不随时间变化
 $\Delta\xi = 1/160$, 节点数不变

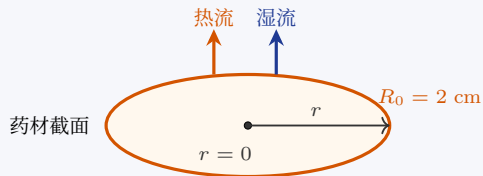


$\xi \in [0, 1]$, 不随时间变化
 $\Delta\xi = 1/160$, 节点数不变

$$R(t) : 2.000 \rightarrow 1.200 \text{ cm} \quad | \quad N = 160 \quad (\Delta\xi = 1/160) \quad | \quad R(t)$$

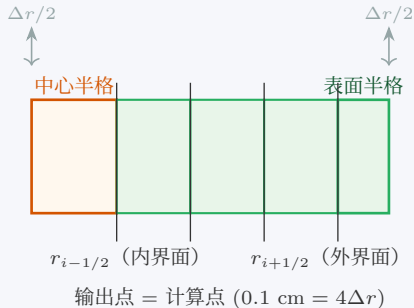
(a) 圆柱截面与边界条件

$$-k\partial_r T = h(T_R - T_\infty), \quad -D\partial_r C = k_m(C_R - C_\infty)$$

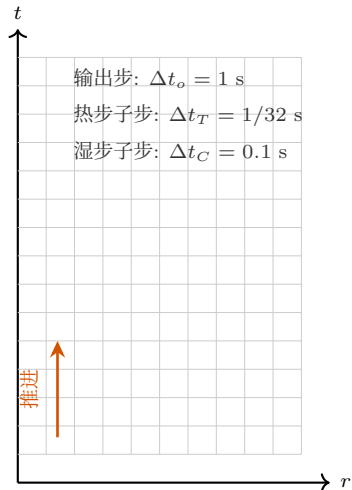


$r = 0$: 对称条件 $\partial_r T = \partial_r C = 0$

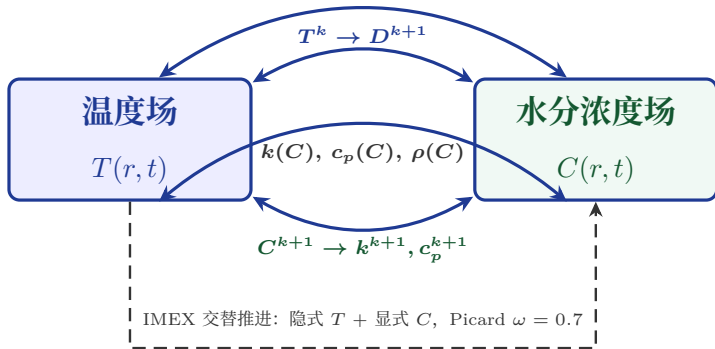
(b) 径向控制体 ($N = 80, \Delta r = 0.25 \text{ mm}$)



(c) 时空离散与推进



$$D = D_0 \exp(-E_a/RT)$$



Q1: 单向耦合 ($T \rightarrow C$, 可分开解)

Q2-Q4: 双向闭环 (物性互馈, 须交替迭代)

输入：初值 + 边界（附件 1）+ 尺寸（附件 2）

空间离散（ $N = 80$ ，元体平衡二阶）

单步隐式推进（追赶法求解）

循环回路未达判据

中心含水率
 $C(0) < 0.15?$

否

步进下一输出步
（自适应/固定步）

是

回退至该输出步起点
保留子步历史

二分回溯

子步内二分回溯
（精度 10^{-3} s）

输出 t_{end} 写 result3.xlsx